L04 — study notes

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Lecture 4 — Multistage games and bargaining · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture. Reconcile with the slides when Aula/Lecture04.pdf arrives (/notes 04 –reconcile).

One claim. Later play can discipline earlier play, but only through equilibria. In a multistage game a threat is credible only if it is a Nash equilibrium of the stage where it is carried out, so behaviour that is not a stage-1 equilibrium needs **a carrot and a stick in a later stage and enough patience** to make the gap bite. **Bargaining is the same**

logic run backwards from a deadline: each proposer offers the responder exactly what rejecting is worth, so agreement is immediate and the split is set by the threat that is never carried out. **Without a deadline, stationarity replaces the last period:** the proposer gets $\frac{1}{1+\delta}$, and **patience is bargaining power**.

Three layers per topic:

Layer	What it means here
U — intuition	Tadelis's prose and examples (lawmakers, gangs, the fisherman); Gibbons's applications (bank runs, tariffs, tournaments); the ultimatum experiments; M&R's bargaining over a sale
C — course (examinable, estimated)	Tadelis Ch. 9 Props. 9.1–9.3, Def. 9.1, Thm. 9.1; Ch. 11 Props. 11.1–11.3, eqs. (11.1)–(11.13); Gibbons's two-stage method and the three-period model
G — formal	MWG Examples 9.AA.1–9.AA.2 (the bound-squeeze uniqueness proof, pie v); the one-stage deviation principle for infinite horizons. **Named or proved, not examinable unless the lecture brings it**

Reading order

Part I — multistage games (Tadelis Ch. 9, Gibbons §2.2)

1. L04-01-multistage-games — §9.1–9.3, pp. 175–180; Gibbons §2.2.A: the definition, the discounted sum, **strategies as history-contingent plans** (32 per player, 1024 profiles in the Prisoner-Revenge Game), and Gibbons's "stage 2 first" method.
2. L04-02-carrot-and-stick — §9.4, pp. 180–184: **Propositions 9.1–9.3**, the carrot and the stick, (M, m) sustained iff $\delta \geq \frac{1}{3}$, (F, m) iff $\delta \geq \frac{2}{3}$, and **the two requirements**.
3. L04-03-one-stage-deviation — §9.5, pp. 184–186: **Theorem 9.1** proved step by step, and the five-line check that replaces comparing 32 strategies.
4. L04-04-two-stage-applications — Gibbons §2.2.B–D, pp. 73–82: **bank runs** (two equilibria, one a run), **tariffs** (a dominant $\frac{a-c}{3}$ and a Prisoners' Dilemma), **tournaments** (the prize that buys efficient effort).

Part II — strategic bargaining (Tadelis Ch. 11, Gibbons §2.1.D, MWG 9.AA)

5. L04-05-ultimatum-game — Ch. 11 intro and §11.1, pp. 220–224: Nash rationalizes **every** split; subgame perfection gives the proposer **everything**; why the responder accepts at indifference; the experiments.
6. L04-06-finite-alternating-offers — §11.2, pp. 224–227; Gibbons pp. 68–70: the induction table, **(11.1)** $x_1 = \frac{1-(-\delta)^T}{1+\delta}$, **immediate agreement** (Prop. 11.3), last-mover vs first-mover advantage, and **the order of limits**.
7. L04-07-rubinstein-infinite-horizon — §11.3, pp. 228–229; Gibbons pp. 70–71; MWG 9.AA.2: the **Shaked–Sutton** argument, the fixed point, $\frac{1-\delta_2}{1-\delta_1\delta_2}$, and the $\Delta \to 0$ bridge to **Nash bargaining**.

8. L04-08-legislative-bargaining — §11.4, pp. 229–235: Baron–Ferejohn; closed rule $k = 1 - \delta\frac{N-1}{2N}$; open rule $\frac{1}{1+2\delta}$; the risky minimal coalition beats it when $\delta > \sqrt{3} - 1$.
9. L04-09-bargaining-private-information — M&R pp. 140–142: why efficient agreement can be **infeasible** once values are private ($p \geq 1.75$ vs $p \leq 1.25$).

Support

10. L04-10-formulas — the study guide mapped, every proposition and equation, every number.
11. L04-11-pitfalls — 13 confusions and 20 original True/False items.
12. L04-12-python — the reference implementation; reuses `check_part2.py::proposer_share`.

The one result to hold

$$\boxed{x^* = \frac{1-\delta_2}{1-\delta_1\delta_2} \xrightarrow{\delta_1=\delta_2=\delta} \frac{1}{1+\delta} \xrightarrow{\delta\to 1} \frac{1}{2}}$$

The proposer's share is set by what the responder could get by waiting. The same sentence gives the finite game's induction, the infinite game's stationarity, the legislative proposer's price per partner, and, in Part I, the carrot-and-stick threshold $\delta \geq g/(c-s)$.

Numbers and cases to have memorized

Case	Value	Meaning
Prisoner-Revenge, (M, m)	$\delta \geq \frac{1}{3}$	gain 1 against a stick worth 3
Prisoner-Revenge, (F, m)	$\delta \geq \frac{2}{3}$	gain 2 against the same stick
Ultimatum SPE	$(1, 0)$	Nash allows any split
Two rounds	$(1-\delta,\ \delta)$	the last word is worth δ
Finite, $\delta = 0.6$	$1,\ 0.4,\ 0.76,\ 0.544, \ldots \to 0.625$	zig-zag to $\frac{1}{1+\delta}$
Rubinstein	$\frac{1}{1+\delta}$; $\frac{1-\delta_2}{1-\delta_1\delta_2}$	first-mover edge only
Tariffs	$t^* = \frac{a-c}{3}$	dominant, and jointly worse than free trade
Legislative, $N = 3$	closed $1 - \frac{\delta}{3}$; open $\frac{1}{1+2\delta}$; switch at $\sqrt{3} - 1$	protocol is power

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Figures

Built by `figures/make_figures.py` (run: `python Notas/L04/figures/make_figures.py`; the script ends with `assert` checks on every figure's key number).

Figure	Shows	Lives in
L04-prisoner-revenge-reduced	the two stage-games and the stage-1 game with continuation payoffs at $\delta = 0.5$	L04-02-carrot-and-stick §3.1

Figure	Shows	Lives in
L04-carrot-stick-threshold	stay vs fink against δ; crossings at $\frac{1}{3}$ and $\frac{2}{3}$	L04-02-carrot-and-stick §3
L04-tariff-dilemma	$W_i^*(t_i)$ for two t_j: same peak, lower level	L04-04-two-stage-applications §2.2
L04-finite-horizon-shares	$x_1(T)$ for $\delta = 0.6, 0.9$ converging to $\frac{1}{1+\delta}$	L04-06-finite-alternating-offers §3.1
L04-shaked-sutton-fixed-point	$f(s) = 1 - \delta + \delta^2 s$ against the diagonal	L04-07-rubinstein-infinite-horizon §2.1
L04-rubinstein-patience	$\frac{1-\delta_2}{1-\delta_1\delta_2}$ against δ_2	L04-07-rubinstein-infinite-horizon §4
L04-legislative-shares	closed, open-guaranteed and open-risky payoffs against δ	L04-08-legislative-bargaining §4

Interactive companions

- The Patience Ledger (`companions/companion-rubinstein-bargaining.html`) — δ_1, δ_2, the horizon T (or no deadline) and the pie v; read-outs for player 1's share (the answer), both payoffs in dollars, the Rubinstein limit, the gap to it, the first-mover edge and the last mover; panels for the share against T and the backward-induction ladder. Linked from notes 06 and 07.
- The Carrot and the Stick (`companions/companion-carrot-and-stick.html`) — δ, the temptation, the sucker payoff and the stick of the Prisoner-Revenge Game; read-outs for both thresholds and the set of sustainable stage-1 outcomes; exact $\delta = \frac{1}{3}$ and $\frac{2}{3}$ buttons. Linked from note 02.
- Both verified by `companions/check_companions.py` (page JavaScript under `node` against Python, three control vectors each).

Scope and ceiling

- **Examinable (estimated):** Tadelis Ch. 9 and §11.1–11.3 in full; Gibbons §2.1.D and §2.2 as the method and its applications. §11.4 (legislative bargaining) is inside the study guide's pages but its weight is unknown until the slides arrive.
- **Named only:** MWG's proofs (G layer); the infinite-horizon one-stage deviation principle (proved for Lecture 5); Baron–Ferejohn's equilibrium multiplicity (Herrero); Nash's axioms.
- **Out** (`CLAUDE.md` scope): mechanism design and the revelation principle (named in note 09 only to place M&R's result); moral hazard (the tournament is used as a two-stage game only); cheap talk.
- **Exercises are cited by number and page only**, never reproduced.

Course-wide position

Lecture	Relation to this one
Lecture 1	strategies as complete contingent plans; dominance (fink, the dominant tariff)
Lecture 2	stage-game Nash equilibria, pure and mixed (the Revenge Game's $q = \frac{1}{2}$); Cournot inside the tariff game

Lecture	Relation to this one
Lecture 3	backward induction and subgame perfection, here applied to stages and to offers
Lecture 4	**multistage games, the carrot and the stick, alternating-offer bargaining**
Lecture 5	the same stage-game forever: no last stage, so cooperation without a second equilibrium; the one-stage deviation principle with discounting
Lecture 6	collusion and relational contracts: carrot-and-stick incentives inside organizations
Lecture 7	private information (note 09's example, adverse selection)
Lecture 8	perfect Bayesian equilibrium; bargaining with delay under asymmetric information

Rules packs: rules/02_barganha (bargaining), rules/game-theory.

Bibliography actually used in these notes

Book	Where
Tadelis (2013) Ch. 9 (pp. 175–189) and Ch. 11 (pp. 220–237) — the spine	notes 01–03, 05–08
Gibbons (1992) §2.1.D (pp. 68–71), §2.2 (pp. 71–82)	notes 01, 04, 06, 07
MWG (1995) Ch. 9 App. A, Examples 9.AA.1–9.AA.2 (pp. 296–298)	the G layer of notes 05–07
Milgrom & Roberts (1992) Ch. 5, pp. 140–142	note 09

Related project material.

- `Aula/aula-02-notes.md` §5 (Nash bargaining; alternating offers, preview) and `Aula/notes-aulas-01-02/main.pdf` part 2, with its checks `Aula/notes-aulas-01-02/check_part2.py`.
- `Listas/lista-aulas-01-02.pdf` Block 3 (Questions 5–6: alternating offers over a Coasean surplus) and `Listas/lista-aulas-01-02-check.py` (`finite_share`).
- `Listas/pratica-aula-01-02.md` items 20, 28, 29 (bargaining practice by number and page).
- `Simulados/quiz-aula02-3-nash-offers-2026-10-13.md`.
- **After-class drill (study guide):** Muñoz-Garcia Ch. 4 Ex. 1 (p. 154), Ex. 16–18 (pp. 198–200); Ch. 6 Ex. 8 (p. 303); MWG 9.B.12, 9.B.13 (pp. 302–303); Tadelis 9.1–9.6 (pp. 186–189), 11.1–11.7 (pp. 236–237).
- Study guide: Map/study-guide (Lecture 4).

9.1–9.3 · A multistage game is one extensive form: payoffs, histories, strategies

> **Claim.** Playing one game after another, with each outcome observed before the next stage, is **one** extensive-form game. Payoffs add up with discounting, and a strategy is a plan that names an action **for every history**. That richer strategy space lets later play be made to depend on earlier play, and that is where the economics of the chapter comes from.

Prev: — · Next: L04-02-carrot-and-stick · Map: L04-00-map

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1 · U — why bother with stages

Tadelis opens with two firms that compete in one market and later meet in another, and with lawmakers who sit on one committee and then, if re-elected, on the next (p. 175). Two questions follow, and the chapter answers both:

Question (p. 175)	Answer (pp. 175–176)
Should rational, forward-looking players treat the sequence as **one grand game**?	**Yes.** They anticipate future games and use them.
Must they play a **stage-game Nash equilibrium in every stage**?	**No.** If future behaviour can be conditioned on past outcomes, they can create **credible, self-enforcing incentives** for early-stage behaviour that is not a stage-game equilibrium.

> **The core insight of multistage games (p. 176): if players can condition future behaviour on past outcomes, the set of self-enforcing outcomes becomes richer.**

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2 · C — the definition (§9.1, p. 176)

A **multistage game** is a **finite** sequence of normal-form **stage-games**, each a simultaneous-move game of complete but imperfect information, played by the **same players** in periods $t = 1, 2, \ldots, T$. Two conventions:

1. after each stage **all players observe its outcome**, and this is common knowledge;
2. total payoffs are evaluated from the **sequence** of stage outcomes (§2 below).

Infinite sequences of the **same** stage-game are repeated games, Chapter 10 (Lecture 5); games in which some stages are played before earlier outcomes are revealed are outside the text (footnotes 1–2, p. 176).

Gibbons's version (§2.2.A, p. 72). Players 1 and 2 choose a_1, a_2 simultaneously; players 3 and 4 observe (a_1, a_2) and then choose a_3, a_4 simultaneously; payoffs are $u_i(a_1, a_2, a_3, a_4)$. Players 3 and 4 may be players 1 and 2 again (a multistage game in Tadelis's sense), or different people (the boss and the workers of a tournament, L04-04-two-stage-applications §3).

2.1 · The running example: the Prisoner-Revenge Game (pp. 176–177)

Stage 1 ($t = 1$), the Prisoner's Dilemma, mum (M/m) or fink (F/f), uppercase for player 1:

	m	f
M	4, 4	-1, 5
F	5, -1	**1**, **1**

Stage 2 ($t = 2$), the Revenge Game, loner (L/l) or join a gang (G/g): two loners get 0 each; two gang members fight, -3 each; a loner facing a gang member gets -4 and the gang member -1:

	l	g
L	**0**, **0**	-4, -1
G	-1, -4	$\mathbf{-3}$, $\mathbf{-3}$

Each stage on its own.

- Stage 1: F strictly dominates M ($5 > 4$, $1 > -1$), so the unique NE is (F, f).
- Stage 2: two pure NE. (L, l): against l, L pays $0 > -1$. (G, g): against g, G pays $-3 > -4$. And a mixed NE. **Worked:** let player 2 play l with probability q. Player 1 is indifferent when

$$u_1(L) = 0 \cdot q + (-4)(1-q) = 4q - 4, \qquad u_1(G) = -1 \cdot q + (-3)(1-q) = 2q - 3,$$

$$4q - 4 = 2q - 3 \iff q = \tfrac{1}{2}.$$

  By symmetry player 1 also plays L with probability $\frac{1}{2}$ — Tadelis's "(verify this!)", p. 177. Each then expects $4(\frac{1}{2}) - 4 = -2$.

  The two pure equilibria of stage 2 are what makes this game interesting: a friendly one worth 0 and a gang war worth -3. Note 02 turns the gap between them into a carrot and a stick.

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3 · C — payoffs: the discounted sum (§9.2, p. 177)

With v_{it} player i's payoff from the stage-t outcome and a common discount factor $\delta \in [0, 1]$,

$$\boxed{v_i = v_{i1} + \delta v_{i2} + \delta^2 v_{i3} + \cdots + \delta^{T-1} v_{iT} = \sum_{t=1}^{T} \delta^{t-1} v_{it}}$$

A higher δ means more patient players; $\delta = 0$ means only the current stage matters.

Worked (p. 177). Play (F, m) then (L, g):

$$v_1 = 5 + \delta(-4), \qquad v_2 = -1 + \delta(-1).$$

With $\delta = \frac{1}{2}$: $v_1 = 5 - 2 = 3$ and $v_2 = -1 - 0.5 = -1.5$.

Figure 9.1 (p. 178) draws the whole game as one tree whose 16 terminal nodes carry these discounted sums. That the payoffs "arrive" only at the end of the tree is harmless: strategically only the sum matters (p. 177).

Notation. Tadelis writes v_i for total payoffs in multistage games, while the Lecture 1 slides write u_i for the payoff function of a normal-form game $G = \langle I, \{S_i\}, \{u_i\}\rangle$. These notes keep Tadelis's v_i for the discounted total and v_{it} for the stage payoff. Reconcile with the slides when they arrive.

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4 · C — strategies condition on histories (§9.3, pp. 178–180)

Because each stage's outcome is revealed before the next, **the number of information sets of a player at stage t equals the number of possible outcomes of stages** $1, \dots, t-1$ (p. 178). Inside a stage the players still move without seeing each other, so each information set holds several nodes.

Prisoner-Revenge (p. 179). Stage 1 has four outcomes Mm, Mf, Fm, Ff, so each player has **four** stage-2 information sets. A pure strategy is a quintuple

$$s_i = (s_{i1},\ s_{i2}(Mm),\ s_{i2}(Mf),\ s_{i2}(Fm),\ s_{i2}(Ff)).$$

Counting. Each of the five entries takes one of two values, so

$$|S_i| = 2^5 = 32, \qquad |S_1 \times S_2| = 32 \times 32 = 1024$$

pure-strategy profiles (p. 179). That is why the extensive form, not a 32×32 matrix, is the tool.

In general (p. 179). A pure strategy is

$$s_i = \{s_{i1},\ s_{i2}(h_1),\ \dots,\ s_{it}(h_{t-1}),\ \dots,\ s_{iT}(h_{T-1})\},$$

where $h_{t-1} \in H_{t-1}$ is a **history**, the sequence of outcomes before t. A behavioural strategy replaces each action by a randomization $\sigma_{it}(h_{t-1})$ (p. 179).

Worked: n firms pricing in T markets (p. 179). Firm i's pure strategy is a list of prices $p_{i1}, p_{i2}(h_1), \dots, p_{iT}(h_{T-1})$, with $h_{t-1} = ((p_{11}, \dots, p_{n1}), \dots, (p_{1,t-1}, \dots, p_{n,t-1}))$. **Each entry after the first is a function from a continuum of histories to a price.** Strategy spaces in multistage games are huge even when every stage is simple.

> **A strategy is a complete contingent plan** (the Lecture 1 slides' definition): in a multistage game "complete" means *one action for every history, including histories the plan itself rules out.* Off-path entries are not decoration; note 02 shows they carry the threats.

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5 · C — the solution concept: subgame perfection, worked stage by stage (Gibbons §2.2.A)

Gibbons solves the two-stage class "in the spirit of backwards induction" (pp. 72–73):

1. **Stage 2 first.** For each first-stage outcome (a_1, a_2), find the NE $(a_3^*(a_1, a_2), a_4^*(a_1, a_2))$ of the game that remains. Here the first step is a *game*, not a one-person optimization as in Lecture 3.
2. **Substitute it back.** Players 1 and 2 then play the simultaneous game with payoffs $u_i(a_1, a_2, a_3^*(a_1, a_2), a_4^*(a_1, a_2))$.
3. If that game has NE (a_1^*, a_2^*), the **subgame-perfect outcome** is $(a_1^*, a_2^*, a_3^*(a_1^*, a_2^*), a_4^*(a_1^*, a_2^*))$.

Gibbons assumes stage 2 has a **unique** NE for every (a_1, a_2) (p. 72). Then step 1 is well defined and the method needs no choice. **Tadelis's Prisoner-Revenge Game deliberately breaks that assumption**, and that is the whole of note 02.

> **Why not threats that are not stage-2 equilibria?** Gibbons, p. 73: when play reaches stage 2, at least one of players 3 and 4 will not carry out such a threat, exactly because it is not a NE of the game that remains. **This is subgame perfection (Lecture 3) applied to stages.**

Links

- Next: L04-02-carrot-and-stick (Propositions 9.1–9.3, the carrot and the stick).
- Applications of the two-stage method: L04-04-two-stage-applications.
- Subgame perfection and backward induction: Lecture 3; repeated games (same stage-game, possibly forever): Lecture 5.
- Rules pack: rules/game-theory items 1–2 (specify the game, name the concept).
- **Exercises (number and page only):** Tadelis 9.1 (p. 186), 9.2 (pp. 186–187).

9.4 · Subgame-perfect equilibria of multistage games: the carrot and the stick

> **Claim.** Stringing stage-game equilibria together is always subgame perfect, and if every stage has a unique equilibrium nothing else is. Behaviour that is **not** a stage-1 equilibrium can be sustained only when a later stage has **two equilibria with different payoffs** (a carrot and a stick) **and** players are patient enough that the gap outweighs the one-shot gain from deviating. In the Prisoner-Revenge Game: mutual mum needs $\delta \geq \frac{1}{3}$; the lopsided (F, m) needs $\delta \geq \frac{2}{3}$.

Prev: L04-01-multistage-games · Next: L04-03-one-stage-deviation · Map: L04-00-map

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1 · C — Proposition 9.1: stage equilibria in sequence are always an SPE (p. 180)

> **Proposition 9.1.** Consider a multistage game with T stages, and let σ^{t*} be a Nash equilibrium strategy profile for the t-th stage-game. There exists a subgame-perfect equilibrium in the multistage game in which the equilibrium path coincides with the path generated by $\sigma^{1*}, \sigma^{2*}, \ldots, \sigma^{T*}$.

Proof, step by step (p. 180). Let every player play σ_i^{t*} at stage t **regardless of history** ("unconditional" strategies).

1. In any subgame starting at stage T: the profile is σ^{T*}, a NE of stage T, and earlier payoffs are sunk constants. So it is a NE of that subgame.
2. In any subgame starting at stage $T-1$: a player's total payoff is the stage-$(T-1)$ payoff **plus a constant** made of (a) the sunk payoffs of earlier stages and (b) the discounted stage-T payoff, which **does not depend on the stage-$(T-1)$ outcome** because stage-T play is unconditional. Adding a constant changes no best response (Tadelis's footnote 3, p. 181), so σ^{T-1*} followed by σ^{T*} is a NE of the subgame.
3. Induct backwards to $t = 1$. ■

Prisoner-Revenge (pp. 180–181). Two pure SPE with unconditional play:

$$s_1 = (F, L, L, L, L),\ s_2 = (f, l, l, l, l) \qquad \text{and} \qquad s_1 = (F, G, G, G, G),\ s_2 = (f, g, g, g, g),$$

and a third one with the mixed Revenge equilibrium after every history (footnote 4, p. 181). In each, stage 2 adds the same constant to every stage-1 cell (0, -3δ or -2δ), so F stays dominant.

> **Unconditional strategies remove every strategic link between the stages.** Each stage is then a separate game, and Proposition 9.1 is just that observation made rigorous.

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2 · C — Propositions 9.2 and 9.3: the last stage pins everything when equilibria are unique

> **Proposition 9.2 (p. 181).** If σ^* is a Nash equilibrium of the multistage game consisting of stage-games $G_1, \ldots, G_T$, then the restriction of σ^* to the stage-game in period T must be a Nash equilibrium of that stage-game.

Why: nothing follows stage T, so the only thing that determines what is best at T is a best response to what others do at T.

> **Proposition 9.3 (p. 181).** If a finite multistage game consists of stage-games that each have a unique Nash equilibrium, then the multistage game has a unique subgame-perfect equilibrium.

Proof by backward induction (p. 182). At T the unique stage NE is played after every history. At $T-1$, stage-T play cannot depend on the stage-$(T-1)$ outcome, so the unique stage-$(T-1)$ NE is played; and so on to $t=1$. ■

> **Corollary to carry to Lecture 5:** a Prisoner's Dilemma repeated a **known finite** number of times has a unique SPE, fink in every period, because each stage has a unique NE. Cooperation needs either a stage with multiple equilibria (this note) or an infinite horizon (Lecture 5).

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3 · C — supporting (M, m) in stage 1 (pp. 182–183)

Target path: (M, m), then (L, l). Proposition 9.2 says stage 2 must be a Revenge NE after **every** history, so the only tool is **which** NE follows which history.

Strategies (p. 182).

$$s_1 = (s_{11}, s_{12}(Mm), s_{12}(Mf), s_{12}(Fm), s_{12}(Ff)) = (M,\ L,\ G,\ G,\ G)$$
$$s_2 = (s_{21}, s_{22}(Mm), s_{22}(Mf), s_{22}(Fm), s_{22}(Ff)) = (m,\ l,\ g,\ g,\ g)$$

"Friendly equilibrium if we both stayed mum; gang war after anything else."

Check, subgame by subgame.

1. *Stage-2 subgames.* After Mm they play (L, l); after Mf, Fm, Ff they play (G, g). Both are NE of the Revenge Game (note 01 §2.1). ✓
2. *The whole game.* Only stage-1 deviations remain to check. Player 1, given s_2:

   $$v_1(M, s_2) = 4 + 0 \cdot \delta, \qquad v_1(F, s_2) = 5 + (-3)\delta.$$

   M is a best response iff

   $$4 \geq 5 - 3\delta \iff 3\delta \geq 1 \iff \boxed{\delta \geq \tfrac{1}{3}} \quad \text{(p. 183)}.$$

   Player 2 is symmetric. ✓

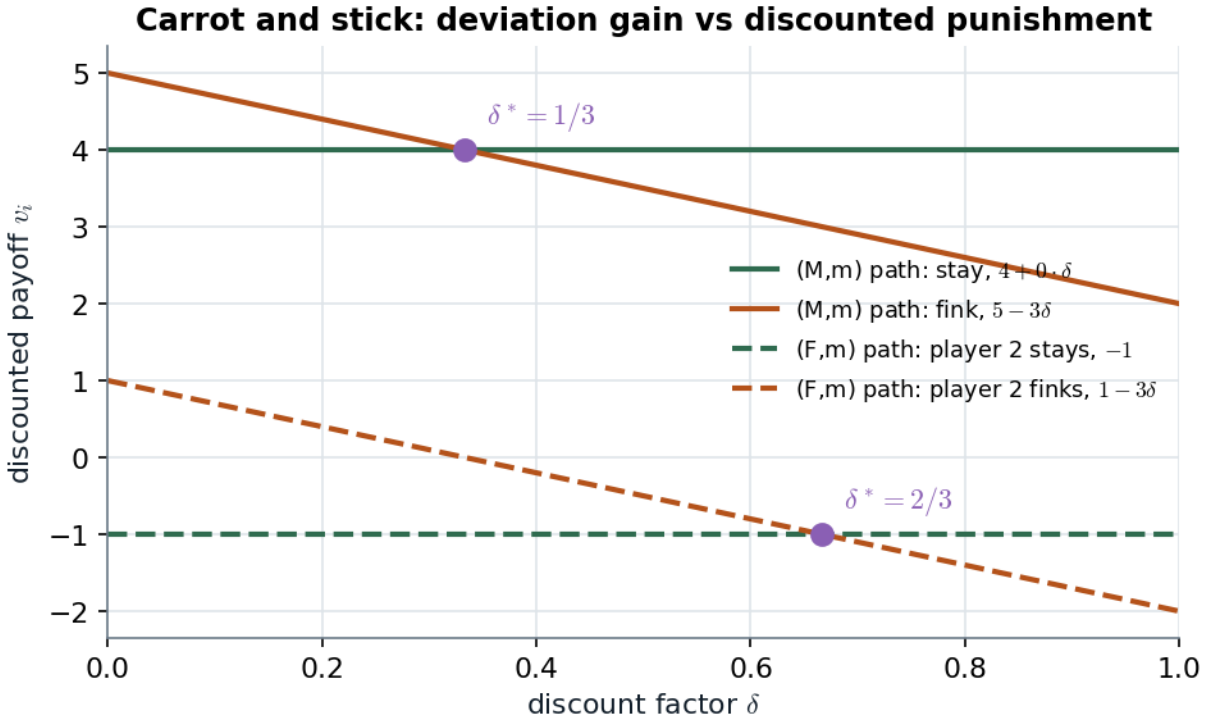


Figure — the deviation gain is fixed (1 for (M,m), 2 for (F,m)); the discounted punishment 3δ grows with patience and overtakes it at $\delta^* = \frac{1}{3}$ and $\frac{2}{3}$. Source: Tadelis pp. 183–184.

Read each pair of lines as "stay" (flat: the carrot pays 0, so patience adds nothing on path) against "fink" (falling: the deviation is followed by the stick, -3, weighted by δ). Cooperation is sustained to the right of each crossing.

3.1 · The reduced stage-1 game

Substitute the continuation values into the stage-1 matrix: $+\delta \cdot 0$ in cell Mm, $+\delta(-3)$ in every other cell. With $\delta = \frac{1}{2}$:

	m	f
M	$4 + 0 = \mathbf{4},\ \mathbf{4}$	$-1 - 1.5 = -2.5,\ 5 - 1.5 = 3.5$
F	$5 - 1.5 = 3.5,\ -2.5$	$1 - 1.5 = -0.5,\ -0.5$

(M, m) is now a NE ($4 > 3.5$ for each), and so is (F, f) ($-0.5 > -2.5$).

Stage 1: Prisoner's Dilemma

	m	f
M	4, 4	-1, 5
F	5, -1	1, 1

Stage 2: Revenge Game (two pure NE)

	l	g
L	0, 0	-4, -1
G	-1, -4	-3, -3

Stage 1 with continuation, $\delta = 0.5$

	m	f
M	4, 4	-2.5, 3.5
F	3.5, -2.5	-0.5, -0.5

Figure — left and centre: the two stage-games (pp. 176–177); right: stage 1 after adding $\delta\times$(continuation) under the carrot-and-stick strategies, $\delta = 0.5$. Source: Tadelis p. 182.

The right-hand table is how to *see* the construction: the linkage subtracts 3δ from every cell except the target, and once 3δ exceeds the one-shot gain of 1 the target becomes a best response.

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4 · C — the two requirements (p. 183)

> **1. There must be at least two distinct equilibria in the second stage: a "stick" and a "carrot." 2. The discount factor has to be large enough for the difference in payoffs between the "stick" and the "carrot" to have enough impact in the first stage.**

Why each is necessary.

- *Requirement 1.* If the target is not a stage-1 NE, some player gains in the short run by deviating. A **credible** punishment must itself be a stage-2 NE (Proposition 9.2), so the only credible punishment is a *switch from a better NE to a worse one*. With one stage-2 NE there is nothing to switch between (that is Proposition 9.3).
- *Requirement 2.* The punishment is worth $\delta \times (\text{carrot} - \text{stick})$ today. A smaller δ "makes the carrot smaller and the stick less painful" (p. 183).

The general threshold. With one-shot deviation gain g and continuation gap $c - s$ (carrot minus stick),

$$\boxed{\text{target sustainable} \iff g \le \delta\,(c-s) \iff \delta \ge \delta^* = \frac{g}{c-s}}$$

For (M, m): $g = 5 - 4 = 1$, $c - s = 0 - (-3) = 3$, $\delta^* = \frac{1}{3}$.

---

5 · Worked — supporting (F, m), the lopsided path (pp. 183–184)

Target: (F, m), then (L, l). **Strategies:**

$$s_1 = (F,\ G,\ G,\ L,\ G), \qquad s_2 = (m,\ g,\ g,\ l,\ g)$$

(friendly only after Fm).

- *Player 1* plays F, a stage-1 dominant strategy, **and** gets the best stage-2 equilibrium. No incentive to deviate.
- *Player 2*:

$$v_2(s_1, m) = -1 + 0 \cdot \delta, \qquad v_2(s_1, f) = 1 + (-3)\delta,$$

$$m \text{ is a best response} \iff -1 \ge 1 - 3\delta \iff \boxed{\delta \ge \tfrac{2}{3}} \quad \text{(p. 184)}.$$

Path	Who is tempted	One-shot gain g	Punishment gap $c - s$	δ^*
(M, m)	both, symmetric	$5 - 4 = 1$	3	**1/3**

Path	Who is tempted	One-shot gain g	Punishment gap $c - s$	δ^*
(F, m)	player 2	$1 - (-1) = 2$	3	**2/3**
(M, f)	player 1 (mirror)	2	3	2/3
(F, f)	nobody	0	—	any δ (Prop. 9.1)

The same stick buys more the smaller the temptation it must overcome. That is why the efficient path (M, m) is *easier* to sustain than (F, m), which hurts player 2 in stage 1.

Interactive companion. The Carrot and the Stick (`companions/companion-carrot-and-stick.html`) — move δ, the temptation T, the sucker payoff S and the stick s; watch both thresholds move, and use the "$\delta = 1/3$ exactly" and "$\delta = 2/3$ exactly" buttons to see the verdicts flip at the book's values. Setting $s = 0$ deletes the stick and every non-(F, f) outcome with it.

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6 · U — what the chapter summary asks you to keep (§9.6, p. 186)

Summary point (p. 186)	Where above
Any sequence of stage-game NE is an SPE, **regardless of** δ	§1
Credible threats and promises in later stages give long-term incentives for short-term actions	§3–§5
Their effectiveness depends on patience; the supportable set often depends on δ	§4

Organizational reading. This is the logic of *relational* incentives inside firms: a future interaction with a good and a bad equilibrium (cooperative or hostile teams, a friendly or a litigious supplier relationship) can discipline behaviour today without any contract. The repeated version, with no last stage, is Lecture 6's relational contracts.

Links

- Previous / next: L04-01-multistage-games · L04-03-one-stage-deviation (why checking one stage at a time suffices).
- The same carrot-and-stick logic with an infinite horizon: Lecture 5 (grim trigger), Lecture 6 (collusion, relational contracts).
- Python: L04-12-python §1 (`two_stage_threshold`).
- Pitfalls: L04-11-pitfalls items 1–3.
- Rules pack: rules/game-theory items 3–4 (multiplicity; finite vs infinite horizon).
- **Exercises (number and page only):** Tadelis 9.1 (p. 186), 9.2 (pp. 186–187), 9.3 (pp. 187–188), 9.4 (p. 188), 9.5 (pp. 188–189), 9.6 (p. 189).

9.5 · The one-stage deviation principle: check one information set at a time

> **Claim.** To verify that a profile is subgame perfect in a finite game you never need to test combinations of deviations: if no player gains by changing his action at **one** information set and following his strategy everywhere else, no multi-stage deviation gains either. This is what made note 02's check a two-line inequality, and it is the workhorse of Lecture 5.

Prev: L04-02-carrot-and-stick · Next: L04-04-two-stage-applications · Map: L04-00-map

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1 · U — the worry it removes (p. 184)

In a five-stage game a player might hope to gain by *combining* deviations: fink at stage 2 **and** switch at stage 4. Checking every combination looks daunting. The principle, due to Blackwell (1965) in dynamic programming, says it is unnecessary. Fixing the others' strategies σ_{-i}, player i faces a one-person decision tree, and in a finite decision tree a plan that cannot be improved one node at a time cannot be improved at all (pp. 184–185).

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2 · C — definition and theorem (p. 185)

Fix σ_{-i}. For an information set h_i let $v_i(\sigma_i, h_i)$ be i's expected payoff from h_i onward. σ_i is **optimal** if no σ_i' and h_i give $v_i(\sigma_i', h_i) > v_i(\sigma_i, h_i)$. Let $\hat{\sigma}_i^{a,h_i}$ be σ_i with the action at h_i replaced by $a \in A_i(h_i)$.

> **Definition 9.1.** A strategy σ_i is **one-stage unimprovable** if there is no information set h_i, action $a \in A_i(h_i)$ and corresponding strategy $\hat{\sigma}_i^{a,h_i}$ such that $v_i(\hat{\sigma}_i^{a,h_i}, h_i) > v_i(\sigma_i, h_i)$.
>
> **Theorem 9.1.** A one-stage unimprovable strategy must be optimal.

(The converse, optimal ⇒ one-stage unimprovable, is immediate.)

2.1 · The proof, step by step (pp. 185–186)

1. Suppose σ_i is one-stage unimprovable but not optimal: some σ_i' and h_i^1 have $v_i(\sigma_i', h_i^1) > v_i(\sigma_i, h_i^1)$.
2. Randomization cannot raise the best attainable payoff, so there is a **path** through finitely many of i's information sets $h_i^1, \dots, h_i^n$ (the last a terminal node) that beats σ_i. Hence finitely many one-stage changes, at $h_i^1, \dots, h_i^{n-1}$, beat σ_i.
3. Let $\hat{\sigma}_i^t$ make the improving changes at $h_i^1, \dots, h_i^t$ and follow σ_i elsewhere. Then
$$v_i(\hat{\sigma}_i^{n-1}, h_i^1) > v_i(\sigma_i, h_i^1). \tag{9.1}$$
4. Look at the **last** change. From h_i^{n-1} on, $\hat{\sigma}_i^{n-2}$ coincides with σ_i, and $\hat{\sigma}_i^{n-1}$ is a one-stage deviation from σ_i there. Unimprovability gives $v_i(\hat{\sigma}_i^{n-2}, h_i^{n-1}) = v_i(\sigma_i, h_i^{n-1}) \geq v_i(\hat{\sigma}_i^{n-1}, h_i^{n-1})$. The two strategies agree before h_i^{n-1}, so
$$v_i(\hat{\sigma}_i^{n-2}, h_i^1) \geq v_i(\hat{\sigma}_i^{n-1}, h_i^1). \tag{9.2}$$
5. (9.1) and (9.2) give $v_i(\hat{\sigma}_i^{n-2}, h_i^1) > v_i(\sigma_i, h_i^1)$: the improvement survives with one change fewer. Repeat down to one change:
$$v_i(\hat{\sigma}_i^1, h_i^1) > v_i(\sigma_i, h_i^1), \tag{9.3}$$
but $\hat{\sigma}_i^1$ is a single one-stage deviation, contradicting unimprovability. ■

Intuition (p. 186): look for the *last* profitable deviation in the sequence. In the subgame where it happens, it is a one-stage deviation that makes the player better off. Peeling deviations off from the end is backward induction in disguise.

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3 · C — how to use it

Recipe (closing remark of §9.4, p. 184). To check that (σ_i, σ_{-i}) is an SPE of a finite game: for every player i, given σ_{-i}, show that i has **no single information set** at which changing the action alone pays.

Worked: note 02's carrot and stick. Player 1's information sets are stage 1 and the four stage-2 sets Mm, Mf, Fm, Ff.

Information set	Prescribed	One-stage deviation	Compare	Verdict
stage 2 after Mm	L (2 plays l)	G	0 vs -1	no gain
stage 2 after Mf, Fm, Ff	G (2 plays g)	L	-3 vs -4	no gain
stage 1	M	F, **then follow** s_1 (play G)	4 vs $5-3\delta$	no gain iff $\delta \geq \frac{1}{3}$

Five checks, each one line, instead of comparing s_1 with 31 alternative strategies.

The deviation is evaluated with the player's *own* strategy continuing afterwards. At stage 1 the deviator finks and then still plays G after Fm, because $s_{12}(Fm) = G$. A combined deviation (fink, then L) is covered automatically: by the stage-2 row it is no better.

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4 · G — finiteness, and what comes next

Theorem 9.1 is stated for trees of **finite length**. Tadelis announces (p. 184) that it will be "even more useful" in Chapter 10, where the horizon is infinite. There the principle still holds when payoffs are discounted ($\delta < 1$), because what happens far in the future matters little (continuity at infinity); that extension is stated and used in Lecture 5. Without discounting it can fail: a player may gain only by deviating at infinitely many dates.

Links

- Previous / next: L04-02-carrot-and-stick · L04-04-two-stage-applications.
- Used again in: L04-07-rubinstein-infinite-horizon §3 (checking the Rubinstein strategies) and Lecture 5 (grim trigger).
- Backward induction in finite perfect-information games: Lecture 3.
- Pitfalls: L04-11-pitfalls item 4.

Gibbons 2.2.B–D · Three two-stage games: a run, a tariff war, a tournament

> **Claim.** The two-stage method of note 01 §5 (solve stage 2 for every stage-1 outcome, substitute, solve stage 1) produces three different lessons. **Bank runs:** stage 2 is harmless, but the reduced stage 1 has **two** equilibria, one of them a run. **Tariffs:** the reduced stage 1 is a **Prisoners' Dilemma** with a dominant tariff of $(a-c)/3$. **Tournaments:** the boss moves first and **chooses the stage-2 game** the workers play; the prize spread buys effort, noise makes it expensive.

Prev: L04-03-one-stage-deviation · Next: L04-05-ultimatum-game · Map: L04-00-map

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1 · Bank runs (§2.2.B, pp. 73–75)

Setup. Two investors each deposit D. The bank's project returns $2r$ if liquidated early and $2R$ at maturity, with $D > r > D/2$ and $R > D$; no discounting. Date 1 is before maturity, date 2 after.

Date 1	withdraw	don't
withdraw	$r,\ r$	$D,\ 2r-D$
don't	$2r-D,\ D$	*go to date 2*

Date 2	withdraw	don't
withdraw	$R,\ R$	$2R-D,\ D$
don't	$D,\ 2R-D$	$R,\ R$

Step 1, date 2. $R > D$ implies $2R - D > R$, so "withdraw" strictly dominates: against withdraw $R > D$, against don't $2R - D > R$. Unique NE: both withdraw, (R, R).

Step 2, substitute (Figure 2.2.1, p. 75).

Date 1 reduced	withdraw	don't
withdraw	**r**, **r**	$D,\ 2r-D$
don't	$2r-D,\ D$	**R**, **R**

Step 3. $r < D$ gives $2r - D < r$: against withdraw, withdraw pays $r > 2r - D$. Against don't, don't pays $R > D$. **Two pure NE.**

Worked with illustrative numbers ($D = 100$, $r = 75$, $R = 150$; they satisfy $100 > 75 > 50$ and $150 > 100$):

Date 1 reduced	withdraw	don't
withdraw	**75**, **75**	100, 50
don't	50, 100	**150**, **150**

Two subgame-perfect outcomes (p. 75): a run, (r, r) at date 1, and no run, (R, R) at date 2. Unlike the Prisoners' Dilemma, an efficient equilibrium exists. The model does not predict *when* runs happen; it shows they **can** happen as equilibrium behaviour (Diamond and Dybvig, 1983). The inefficiency is a **coordination failure**, not a dominant-strategy trap.

Note the structure: Gibbons's §2.2.A assumed a unique stage-2 NE; here stage 2 is unique, and the **multiplicity is in stage 1**.

---

2 · Tariffs and imperfect international competition (§2.2.C, pp. 75–79)

Setup. Two identical countries $i = 1, 2$. Inverse demand in country i: $P_i = a - Q_i$. Firm i sells h_i at home and exports e_i, with constant marginal cost c, and pays tariff t_j per unit exported to country j. **Timing:** governments choose t_1, t_2 simultaneously; firms observe them and choose $(h_1, e_1), (h_2, e_2)$ simultaneously. **Payoffs:** firm profit, and government welfare

$$W_i = \underbrace{\tfrac{1}{2}Q_i^2}_{\text{consumer surplus}} + \pi_i + \underbrace{t_i e_j}_{\text{tariff revenue}} .$$

2.1 · Stage 2: Cournot in each market

Firm i's profit splits into a home part and an export part, so it solves two problems:

$$\max_{h_i} \left[a - (h_i + e_j^*) - c\right] h_i \;\Rightarrow\; h_i^* = \tfrac{1}{2}(a - e_j^* - c), \tag{2.2.1}$$

$$\max_{e_i} \left[a - (e_i + h_j^*) - c - t_j\right] e_i \;\Rightarrow\; e_i^* = \tfrac{1}{2}(a - h_j^* - c - t_j). \tag{2.2.2}$$

Market i is a Cournot duopoly between firm i (cost c) and firm j (cost $c + t_i$). Solve the pair $h_i = \frac{1}{2}(a - e_j - c)$, $e_j = \frac{1}{2}(a - h_i - c - t_i)$:

$$h_i = \tfrac{1}{2}\Big(a - c - \tfrac{1}{2}(a - h_i - c - t_i)\Big) \;\Rightarrow\; 4h_i = a - c + h_i + t_i \;\Rightarrow\; h_i^* = \frac{a - c + t_i}{3},$$

$$e_j^* = \tfrac{1}{2}\Big(a - c - t_i - \frac{a - c + t_i}{3}\Big) = \frac{a - c - 2t_i}{3}. \tag{2.2.3}$$

A tariff raises the home firm's output and cuts the import by twice as much (p. 77).

2.2 · Stage 1: the governments' game

Write $A = a - c$. On market i: $Q_i = h_i^* + e_j^* = (2A - t_i)/3$, and the price-cost margins equal the quantities ($P_i - c = h_i^*$, $P_j - c - t_j = e_i^*$), so $\pi_i = h_i^{*2} + e_i^{*2}$. Hence

$$W_i^*(t_i, t_j) = \frac{(2A - t_i)^2}{18} + \frac{(A + t_i)^2}{9} + \frac{(A - 2t_j)^2}{9} + \frac{t_i(A - 2t_i)}{3} \quad \text{(p. 78)}.$$

t_j enters only through the export-profit term, which does not involve t_i. Differentiate:

$$\frac{\partial W_i^*}{\partial t_i} = -\frac{2A - t_i}{9} + \frac{2(A + t_i)}{9} + \frac{A - 4t_i}{3} = \frac{-2A + t_i + 2A + 2t_i + 3A - 12t_i}{9} = \frac{3A - 9t_i}{9} = 0$$

$$\boxed{t_i^* = \frac{a-c}{3} \quad \text{for each } i, \text{ independent of } t_j}$$

(second derivative $(1+2-12)/9 < 0$). **A dominant strategy.** Substituting: $h_i^* = 4(a-c)/9$, $e_i^* = (a-c)/9$, total $5(a-c)/9$ per market, against $2(a-c)/3 = 6(a-c)/9$ under free trade.

The governments' game in numbers (units of $A^2/162$, computed from W_i^*):

	$t_j = 0$	$t_j = A/3$
$t_i = 0$	72, 72	56, 81
$t_i = A/3$	81, 56	**65, 65**

Check one cell: $W_i^*(A/3, A/3) = \frac{(5/3)^2}{18} + \frac{(4/3)^2}{9} + \frac{(1/3)^2}{9} + \frac{(1/3)(1/3)}{3} = \frac{25}{162} + \frac{32}{162} + \frac{2}{162} + \frac{6}{162} = \frac{65}{162}$.

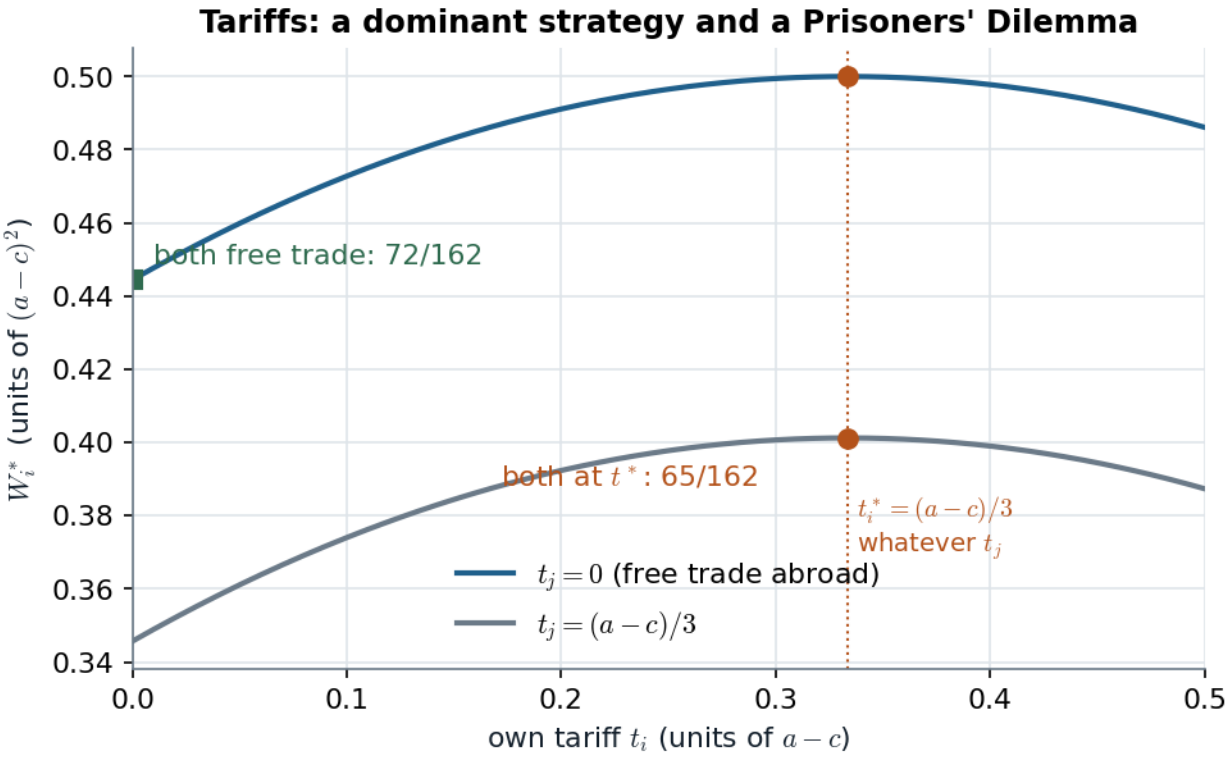


Figure — $W_i^*(t_i, t_j)$ with $a - c = 1$: the peak sits at $t_i = 1/3$ whatever t_j (dominance), yet both at t^* (65/162) is worse than both free (72/162). Source: Gibbons p. 78.

The foreign tariff only shifts the curve down (it taxes *our* exports), never moves its peak; that is what "dominant" looks like. Free trade maximizes joint welfare among non-negative tariffs, so the governments face a **Prisoners' Dilemma** and have a reason to sign a free-trade treaty (pp. 78–79). (If subsidies were allowed the joint optimum would be $t = -(a-c)$, p. 78.)

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3 · Tournaments (§2.2.D, pp. 79–82)

Setup. Worker $i \in \{1, 2\}$ produces $y_i = e_i + \varepsilon_i$, effort $e_i \geq 0$, noise ε_i i.i.d. with zero-mean density f. Output is observed, effort is not. The boss pays w_H to the higher output, w_L to the other. Worker

payoff $w - g(e)$ with $g' > 0$, $g'' > 0$; boss payoff $y_1 + y_2 - w_H - w_L$. In Gibbons's mapping the boss is player 1 (stage 1: choose w_H, w_L), there is no player 2, and the workers are players 3 and 4 (stage 2: choose efforts).

Stage 2. Worker i solves (2.2.4) $\max_{e_i}\ w_L + (w_H - w_L)\Pr\{y_i(e_i) > y_j(e_j^*)\} - g(e_i)$, with first-order condition

$$(w_H - w_L)\,\frac{\partial \Pr\{y_i > y_j\}}{\partial e_i} = g'(e_i). \tag{2.2.5}$$

Conditioning on ε_j:

$$\Pr\{\varepsilon_i > e_j - e_i + \varepsilon_j\} = \int [1 - F(e_j - e_i + \varepsilon_j)] f(\varepsilon_j)\, d\varepsilon_j, \qquad \frac{\partial}{\partial e_i} = \int f(e_j - e_i + \varepsilon_j) f(\varepsilon_j)\, d\varepsilon_j.$$

In a symmetric equilibrium $e_i = e_j = e^*$:

$$(w_H - w_L) \int f(\varepsilon)^2\, d\varepsilon = g'(e^*). \tag{2.2.6}$$

Normal noise, worked. With $\varepsilon \sim N(0, \sigma^2)$, $f(\varepsilon)^2 = \frac{1}{2\pi\sigma^2} e^{-\varepsilon^2/\sigma^2}$ and $\int e^{-\varepsilon^2/\sigma^2} d\varepsilon = \sigma\sqrt{\pi}$, so

$$\int f^2 = \frac{\sigma\sqrt{\pi}}{2\pi\sigma^2} = \frac{1}{2\sigma\sqrt{\pi}},$$

which falls in σ: a bigger prize buys more effort, more noise buys less (p. 81).

Stage 1. Each worker wins with probability $\frac{1}{2}$ in the symmetric equilibrium, so participation requires

$$\tfrac{1}{2} w_H + \tfrac{1}{2} w_L - g(e^*) \geq U_a. \tag{2.2.7}$$

The boss makes it bind: $w_L = 2U_a + 2g(e^*) - w_H$ (2.2.8). Expected profit becomes $2e^* - w_H - w_L = 2e^* - 2U_a - 2g(e^*)$, maximized at $g'(e^*) = 1$; then (2.2.6) gives the **optimal prize**

$$\boxed{(w_H - w_L) \int f(\varepsilon)^2 d\varepsilon = 1} \qquad \text{normal noise: } w_H - w_L = 2\sigma\sqrt{\pi}.$$

Illustrative: $g(e) = e^2/2$ gives $e^* = 1$; with $\sigma = 1$ the spread is $2\sqrt{\pi} \approx 3.54$ (checked in L04-12-python §2).

> **The first mover designs the second stage.** The boss is not in the workers' game; she picks the prize that makes the workers' equilibrium effort equal the efficient one. The incentive question behind it (hidden effort) is the subject of moral hazard, which is outside this course unless a lecture brings it (`CLAUDE.md` scope); here the tournament is used only as a two-stage game.

Links

- The method: L04-01-multistage-games §5. Coordination failures with two equilibria: Lecture 2; Cournot best responses: Lecture 2.
- Python: L04-12-python §2 (tariff welfare table, tournament integral).
- Pitfalls: L04-11-pitfalls item 5.
- **Exercises (number and page only):** Gibbons Problems 2.6–2.9 (p. 133); Tadelis 9.4 (p. 188), 9.6 (p. 189).

11 intro · 11.1 · The ultimatum game: Nash says anything, subgame perfection says everything

> **Claim.** With one round of bargaining, **every** split of the pie is a Nash equilibrium, because the responder can "threaten" to reject small offers. Sequential rationality kills those threats: a responder offered something positive accepts it, so the proposer takes the whole pie. **The gap between the two answers is the case for subgame perfection in bargaining.**

Prev: L04-04-two-stage-applications · Next: L04-06-finite-alternating-offers · Map: L04-00-map

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1 · C — the bargaining protocol (Ståhl, Rubinstein; pp. 220–222)

Two players split a pie of size 1 (it stands for the gains from trade). Every rejection shrinks the pie by the discount factor $0 < \delta < 1$: in period t it is worth δ^{t-1} (p. 221).

Round	Proposer	Offer $(x, 1-x)$	Responder	Accept: payoffs
1	player 1	x to 1, $1-x$ to 2	player 2	$(x,\ 1-x)$
2	player 2	x to 1, $1-x$ to 2	player 1	$(\delta x,\ \delta(1-x))$
t	1 if t odd, 2 if even	same convention	the other	$(\delta^{t-1}x,\ \delta^{t-1}(1-x))$

Convention: x is always *player 1's* share, whoever proposes (Gibbons uses the same convention with s_t, p. 69). If round T ends without agreement, both get 0, a **hard deadline** (Tadelis's fisherman and restaurateur, p. 221).

How it differs from a repeated game (p. 221): the game can **end in any round**, and payoffs arrive **only at agreement**, not as a flow.

---

2 · C — Proposition 11.1: any split is a Nash equilibrium (p. 223)

> **Proposition 11.1.** In the bargaining game, if $T = 1$ then any division of surplus $x^* \in [0, 1]$, $(v_1, v_2) = (x^*, 1 - x^*)$, can be supported as a Nash equilibrium.

Proof. Player 1: "propose x^*". Player 2: "accept any $x \leq x^*$, reject any $x > x^*$".

- Given player 2's rule, any $x > x^*$ yields 0 and any $x < x^*$ yields less than x^*; so x^* is a best response.
- Given x^* is offered, accepting yields $1 - x^* \geq 0$ and rejecting 0; so the rule (which accepts) is a best response. **The rule's rejections are never tested on the path.** ■

Nash equilibrium "has no bite" here (p. 223): it rationalizes every division.

---

3 · C — Proposition 11.2: the unique SPE gives the proposer everything (p. 223)

The flaw in §2. Player 2's rule says "there is a minimum I accept". If offered $1 - x = \varepsilon > 0$ below that minimum, rejecting yields 0 and accepting $\varepsilon > 0$. **Rejecting a positive offer is not sequentially rational.** So in any SPE player 2 accepts every $x < 1$.

Proposition 11.2. The bargaining game with $T = 1$ admits a unique subgame-perfect equilibrium, in which player 1 offers $x = 1$ and player 2 accepts any offer $x \leq 1$.

Proof, both cases.

1. Player 2 accepts every $x < 1$ (above). At $x = 1$ she is indifferent.
2. *If she accepts* $x = 1$*:* player 1's unique best response is $x = 1$. This is the SPE.
3. *If she rejects* $x = 1$ (accepts only $x < 1$): player 1 has **no best response**. Any $x' < 1$ is beaten by $x'' = x' + \frac{1-x'}{2}$, which is still below 1 and still accepted (footnote 1, p. 223); the supremum 1 is not attained. So this strategy cannot be part of an SPE. ■

Concept	Prediction for $T = 1$
Nash equilibrium	any $(x^*, 1 - x^*)$, $x^* \in [0, 1]$
Subgame-perfect equilibrium	$(1, 0)$, unique

The responder's indifference at $x = 1$ is resolved in favour of accepting, not by assumption but because the alternative leaves the proposer without a best response. MWG make the same point (footnote 16, p. 297): with offers on a penny grid the proposer keeps the pie minus one penny, and the outcome converges to $(1, 0)$ as the penny shrinks.

Worked: the same in money (illustrative numbers). A buyer and a seller have gains from trade of \$1,000; the seller makes a take-it-or-leave-it price offer. The SPE price extracts all \$1,000; any price between the seller's cost and the buyer's value is a Nash outcome. The take-it-or-leave-it power is worth the entire surplus.

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4 · U — the experiments (Remark, pp. 223–224)

In laboratory ultimatum games proposers typically offer **25%–50%** of the pie, and responders usually **reject** offers below about **20%–25%** (Camerer, 2003). Responders act with "spite": they prefer nothing to a split that gives the proposer much more.

How to read this. The prediction assumed payoffs equal the money. If the responder's utility also falls with the proposer's advantage, rejecting a small offer can be sequentially rational, and the SPE of *that* game gives the responder a positive share. The game theory is fine; the payoff specification changed. M&R's Food for Thought 5.1 (pp. 163–164) asks the organizational version (fairness and vengeance in commitment), set as practice item 20 in `Listas/pratica-aula-01-02.md`.

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5 · C — the indifference convention, carried forward

Every backward-induction step in notes 06–07 repeats §3: **the responder accepts when indifferent**, and the proposer offers *exactly* the responder's continuation value. Hold that convention fixed; it is what makes the formulas closed-form.

Links

- Next: L04-06-finite-alternating-offers (two rounds and more).

- Subgame perfection and non-credible threats: Lecture 3.
- Earlier treatment: `Aula/aula-02-notes.md` §5.2 ("Ultimatum (one round)").
- Python: L04-12-python §3. Pitfalls: L04-11-pitfalls items 6–7.
- **Exercises (number and page only):** Tadelis 11.1 (p. 236), 11.2 (p. 236); Muñoz-Garcia Ch. 4 Ex. 1 (p. 154).

11.2 · Finitely many rounds: backward induction, the last word, and the first move

> **Claim.** With a deadline, backward induction pins a **unique** SPE: the last proposer takes the pie, and each step back the proposer offers the responder **exactly the discounted value of rejecting**. Agreement is immediate. Player 1's share is $x_1 = \frac{1-(-\delta)^T}{1+\delta}$: it zig-zags with the parity of T (**last-mover advantage**) and settles at $\frac{1}{1+\delta} > \frac{1}{2}$ as $T \to \infty$ (**first-mover advantage** from discounting).

Prev: L04-05-ultimatum-game · Next: L04-07-rubinstein-infinite-horizon · Map: L04-00-map

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1 · C — Nash still says anything, and even "any time" (p. 224)

The strategies of Proposition 11.1 extend to any horizon: in odd rounds 1 proposes x^* and 2 accepts $x \le x^*$; in even rounds 2 proposes x^* and 1 accepts $x \ge x^*$. Worse, Nash cannot even pin **when** agreement happens: let player 1 open with $x = 1$ and player 2 reject everything in round 1; agreement comes in round 2, and the pair is still a NE (p. 224). Subgame perfection restores a unique answer.

---

2 · C — two rounds (Figure 11.2, pp. 224–225)

Work backwards, $\delta < 1$.

1. **Round 2** is an ultimatum game with player 2 proposing: she offers $x = 0$ and keeps the pie, worth δ in round-1 terms.
2. **Round 1, player 2 responds.** Rejecting is worth δ, so she accepts iff $1 - x \ge \delta$, i.e. $x \le 1 - \delta$.
3. **Round 1, player 1 proposes** the largest accepted x:

$$\boxed{x_1 = 1 - \delta, \qquad (v_1, v_2) = (1 - \delta,\ \delta)}$$

With $\delta = 1$ player 2 takes everything (pure last-mover advantage); with $\delta < 1$ player 1 keeps $1 - \delta$ only because waiting costs player 2 (p. 225).

---

3 · C — odd T: the induction written out (pp. 225–226)

T odd, so player 1 proposes first **and** last. Payoffs are in round-1 values.

Round	Proposer	Responder accepts iff	Offer x (player 1's share)	v_1	v_2
T	1	anything (ultimatum)	1	δ^{T-1}	0
$T-1$	2	$\delta^{T-2}x \ge \delta^{T-1}$, i.e. $x \ge \delta$	δ	δ^{T-1}	$\delta^{T-2}(1-\delta)$
$T-2$	1	$\delta^{T-3}(1-x) \ge \delta^{T-2}(1-\delta)$	$1 - \delta + \delta^2$	$\delta^{T-3} - \delta^{T-2} + \delta^{T-1}$	$\delta^{T-2} - \delta^{T-1}$

Round	Proposer	Responder accepts iff	Offer x (player 1's share)	v_1	v_2
$T-3$	2	$\delta^{T-4}x \geq \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$	$\delta - \delta^2 + \delta^3$	as above	$\delta^{T-4} - \delta^{T-3} + \delta^{T-2} - \delta^{T-1}$

Check one step ($T-2$). Divide the acceptance condition by δ^{T-3}: $1 - x \geq \delta(1-\delta)$, so the largest accepted offer is $x = 1 - \delta + \delta^2$. ✓

The pattern (p. 226). In odd rounds $T-s$ (s even) the offer is $1 - \delta + \delta^2 - \cdots + \delta^s$; in even rounds ($s$ odd) it is $\delta - \delta^2 + \cdots + \delta^s$. In round 1, $s = T-1$:

$$x_1 = 1 - \delta + \delta^2 - \delta^3 + \cdots + \delta^{T-1} = \underbrace{(1 + \delta^2 + \cdots + \delta^{T-1})}_{\frac{1-\delta^{T+1}}{1-\delta^2}} - \underbrace{(\delta + \delta^3 + \cdots + \delta^{T-2})}_{\frac{\delta-\delta^T}{1-\delta^2}} = \frac{(1-\delta)(1+\delta^T)}{(1-\delta)(1+\delta)}.$$

$$\boxed{v_1 = x_1 = \frac{1+\delta^T}{1+\delta}, \qquad v_2 = 1 - x_1 = \frac{\delta - \delta^T}{1+\delta} \qquad (T \text{ odd})} \tag{11.1}$$

Even T (MWG Example 9.AA.1, p. 297; Tadelis leaves it as Exercise 11.3). Now player 2 is the first proposer of the odd-length game that starts in round 2, which is worth $\delta \cdot \frac{1+\delta^{T-1}}{1+\delta}$ to her today; player 1 offers exactly that:

$$x_1 = 1 - \delta\,\frac{1+\delta^{T-1}}{1+\delta} = \frac{1-\delta^T}{1+\delta} \qquad (T \text{ even}).$$

Both cases in one line: $x_1(T) = \displaystyle\sum_{k=0}^{T-1} (-\delta)^k = \frac{1-(-\delta)^T}{1+\delta}$.

3.1 · Worked, $\delta = 0.6$ (the project's calibration, `check_part2.py`)

T	1	2	3	4	5	6	∞
x_1	1	$1 - 0.6 = 0.400$	$1 - 0.6 + 0.36 = 0.760$	$\frac{1-0.1296}{1.6} = 0.544$	$\frac{1+0.07776}{1.6} = 0.674$	$\frac{1-0.046656}{1.6} = 0.596$	$\frac{1}{1.6} = 0.625$
last word	1	2	1	2	1	2	—

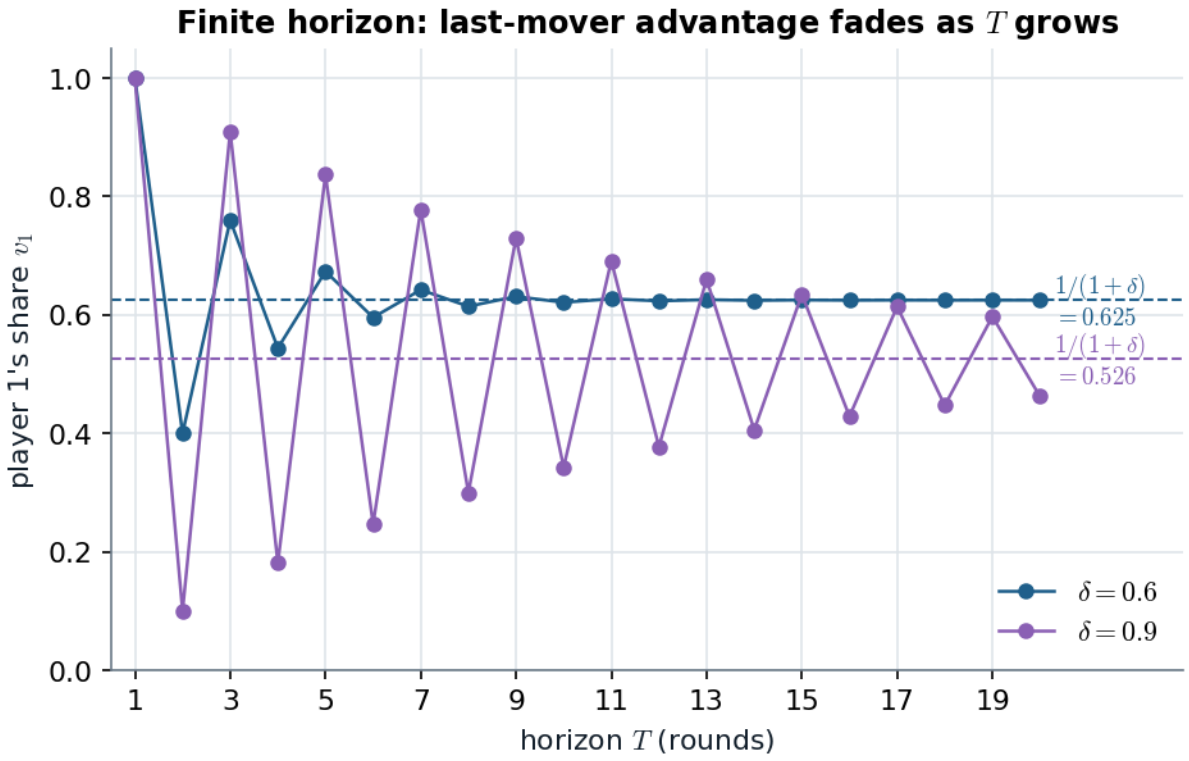


Figure — $x_1(T) = \frac{1-(-\delta)^T}{1+\delta}$ *for* $\delta = 0.6$ *and* 0.9*; dashed lines at* $1/(1+\delta)$*. Source: eq. (11.1), p. 226, and MWG p. 297.*

The distance to the limit is $\delta^T/(1+\delta)$, so the last-mover advantage decays geometrically, and **more slowly the more patient the players**: at $\delta = 0.9$ it is still visible at $T = 20$.

---

4 · C — Proposition 11.3: agreement in round 1 (p. 226)

> **Proposition 11.3.** Any subgame-perfect equilibrium must have the players reach an agreement in the first round.

Proof. Suppose agreement came later with payoffs (v_1, v_2). Discounting wastes part of the pie, so $v_1 + v_2 < 1$. Player 1 could instead offer $x = 1 - v_2 - \varepsilon$ in round 1: player 2 gets $v_2 + \varepsilon$ now, more than by continuing, so sequential rationality makes her accept; and for small ε, player 1 gets $1 - v_2 - \varepsilon > v_1$. A profitable deviation, a contradiction. ■

> **With complete information, delay is pure waste, and rational bargainers do not waste.** The off-path rounds still matter: they set the price of round 1. Where delay *does* occur in reality (strikes), something in this model must be missing: L04-09-bargaining-private-information.

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5 · C — what the formula says about power (pp. 226–227)

Case	x_1	Reading
$\delta = 0$	1 for every T	only round 1 matters: an ultimatum game
$\delta = 1$, T odd	1	no cost of waiting: **only the last word matters**, however long the game
$\delta = 1$, T even	0	the same, for player 2
T odd, $0 < \delta < 1$	$v_1 > v_2$	player 1 has both advantages

The limits (11.2)–(11.3). Fix $\delta \in (0,1)$ and lengthen the game: $\delta^T \to 0$, so

$$\lim_{T\to\infty} v_1 = \frac{1}{1+\delta}, \qquad \lim_{T\to\infty} v_2 = \frac{\delta}{1+\delta}. \tag{11.2}$$

The deadline effect disappears; only the first-mover advantage from discounting remains. Then let players become patient:

$$\lim_{\delta\to 1}\lim_{T\to\infty} v_1 = \lim_{\delta\to 1}\frac{1}{1+\delta} = \frac{1}{2} = \lim_{\delta\to 1}\lim_{T\to\infty} v_2. \tag{11.3}$$

The order of limits matters (footnote 2, p. 227). Take $\delta \to 1$ first, at fixed odd T: $v_1 = \frac{1+1}{1+1} = 1$, and then $T \to \infty$ keeps it at 1. The artificial take-it-or-leave-it advantage survives. **Long horizon first, patience second, gives the equal split.**

Interactive companion. The Patience Ledger (`companions/companion-rubinstein-bargaining.html`) — slide T and watch the highlighted dot jump above and below the dashed limit; the ladder panel shows every round's offer, the path that is never played. The "$\delta = 1$ first, $T = 39$" button is footnote 2.

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6 · Worked — Gibbons's three-period model and different discount factors

Gibbons (pp. 68–70). Round 3's settlement $(s, 1-s)$ is given exogenously. Round 2: player 1 accepts s_2 iff $s_2 \geq \delta s$, so player 2 offers $s_2^* = \delta s$ (keeping $1-\delta s$, better than $\delta(1-s)$). Round 1: player 2 accepts iff $1 - s_1 \geq \delta(1 - s_2^*)$, so

$$s_1^* = 1 - \delta(1-\delta s) = 1 - \delta + \delta^2 s.$$

With $s = 1$ (player 1 would get all in round 3) this is $1 - \delta + \delta^2$, the $T = 3$ value of (11.1). Note 07 reuses $f(s) = 1 - \delta + \delta^2 s$ to solve the infinite game.

Different discount factors (δ_1 for player 1, δ_2 for player 2; three rounds, player 1 proposes in rounds 1 and 3):

1. Round 3: player 1 takes the pie: $s_3 = 1$.
2. Round 2: player 1 accepts iff $s_2 \geq \delta_1 \cdot 1$; player 2 offers $s_2 = \delta_1$.
3. Round 1: player 2 accepts iff $1 - s_1 \geq \delta_2(1-\delta_1)$; player 1 offers

$$s_1 = 1 - \delta_2(1-\delta_1).$$

Numbers ($\delta_1 = 0.9$, $\delta_2 = 0.7$, the values verified in `check_part2.py`): $s_1 = 1 - 0.7 \times 0.1 = 0.93$. This is exactly `Listas/lista-aulas-01-02.tex` Block 3, Question 5a (answer E), there with the joint value W^* as the pie.

6.1 · G — MWG Example 9.AA.1 (pp. 296–297)

MWG run the same induction on a pie of v dollars (Ståhl's model) and get, for odd T, $v_1^*(T) = v\,[1 - \delta + \delta^2 - \cdots + \delta^{T-1}] = v\,\frac{1+\delta^T}{1+\delta}$, with $v_2^*(T) = v - v_1^*(T)$, and the even case as in §3. Scaling the pie scales every payoff and no share.

Links

- Previous / next: L04-05-ultimatum-game · L04-07-rubinstein-infinite-horizon.
- Earlier treatment: `Aula/aula-02-notes.md` §5.2; `Aula/notes-aulas-01-02/main.pdf` part 2 and `check_part2.py::proposer_share` (the same induction, any δ_1, δ_2); `Listas/lista-aulas-01-02-check.py::finite_share` checks (11.1) symbolically for $T \leq 8$.
- Python: L04-12-python §3. Pitfalls: L04-11-pitfalls items 8–10.
- Rules pack: rules/02_barganha ("Alternating offers").
- **Exercises (number and page only):** Tadelis 11.3 (p. 236), 11.4 (p. 236), 11.5 (pp. 236–237); Muñoz-Garcia Ch. 4 Ex. 16 (p. 198), Ex. 17 (p. 199), Ex. 18 (p. 200); MWG 9.B.13 (pp. 302–303); `Listas/lista-aulas-01-02.pdf` Block 3, Question 5a.

11.3 · Rubinstein's infinite horizon: stationarity replaces the last period

> **Claim.** With no deadline there is no last round to start backward induction from, yet the game still has a **unique** SPE. Every odd round looks like round 1 and every even round like round 1 with the roles swapped, so the best and the worst equilibrium payoffs must solve the same pair of equations: the proposer gets $\frac{1}{1+\delta}$, agreement is immediate, and with different patience the share is $\frac{1-\delta_2}{1-\delta_1\delta_2}$. **Patience is bargaining power**, and as offers become frequent the split converges to the Nash bargaining solution with weights set by the discount rates.

Prev: L04-06-finite-alternating-offers · Next: L04-08-legislative-bargaining · Map: L04-00-map

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1 · U — why the finite method breaks, and what replaces it (p. 228)

Disagreeing forever gives 0 (discounting leaves nothing to split), but the path of perpetual disagreement is **infinitely long**, so there is no terminal node to fold back from. What the game has instead is a **stationary structure**: the subgame after any two rejections *is* the original game (Gibbons p. 70, crediting Shaked and Sutton, 1984). The infinite game is **not** "the limit of the finite games" by definition; that the answers coincide is a result, and MWG stress it is not a general property of infinite games (p. 298; Lecture 5's repeated games are the counterexample).

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2 · C — the Shaked–Sutton argument (Tadelis pp. 228–229)

Step 0: immediate agreement. The logic of Proposition 11.3 applies unchanged: delay wastes pie that a slightly better round-1 offer could share. So in every SPE the first offer is accepted.

Step 1: best and worst. Suppose several SPE exist. Let $\bar{v}$ and $\underline{v}$ be the **highest and lowest** SPE payoffs of the player who proposes in round 1. By stationarity the same two numbers bound player 2's payoff in the subgame where **she** proposes (round 2), and vice versa.

Step 2: the worst for player 1 is the best for player 2. In the SPE worst for player 1, player 2 can secure $\bar{v}$ by rejecting and proposing in round 2, worth $\delta\bar{v}$ today. Player 1 must offer at least that, and offers no more:

$$\underline{v} = 1 - \delta\bar{v}. \tag{11.4}$$

Step 3: symmetrically, in the SPE best for player 1, player 2's threat is worth $\delta\underline{v}$:

$$\bar{v} = 1 - \delta\underline{v}. \tag{11.5}$$

Step 4: solve. Subtract (11.4) from (11.5): $\bar{v} - \underline{v} = \delta(\bar{v} - \underline{v})$, so $(1-\delta)(\bar{v} - \underline{v}) = 0$ and, since $\delta < 1$, $\bar{v} = \underline{v} \equiv v$. Then $v = 1 - \delta v$:

$$\boxed{v_1^* = \frac{1}{1+\delta}, \qquad v_2^* = \frac{\delta}{1+\delta}}$$

The strategies (p. 229). In odd rounds player 1 offers $x = \frac{1}{1+\delta}$ and player 2 accepts any $x \leq \frac{1}{1+\delta}$. In even rounds player 2 offers $x = \frac{\delta}{1+\delta}$ (player 1's share) and player 1 accepts any $x \geq \frac{\delta}{1+\delta}$.

Worked, $\delta = 0.6$: $v_1^* = 1/1.6 = 0.625$, $v_2^* = 0.6/1.6 = 0.375$; with $\delta = 0.9$: 0.526 against 0.474. This is the $T \to \infty$ limit (11.2) of note 06, now derived without any deadline.

2.1 · Gibbons's version: a fixed point (pp. 70–71)

Gibbons truncates the game after round 2: if the continuation from round 3 gives player 1 the payoff s, the three-period logic of note 06 §6 gives player 1, in round 1,

$$f(s) = 1 - \delta(1 - \delta s) = 1 - \delta + \delta^2 s.$$

Let s_H be the highest backward-induction payoff of player 1 in the whole game. Using s_H as the round-3 payoff yields $f(s_H)$, the highest possible round-1 payoff because f is increasing; but that *is* s_H, so $f(s_H) = s_H$. Likewise $f(s_L) = s_L$. The only fixed point:

$$s = 1 - \delta + \delta^2 s \iff s(1 - \delta^2) = 1 - \delta \iff s^* = \frac{1}{1+\delta}.$$

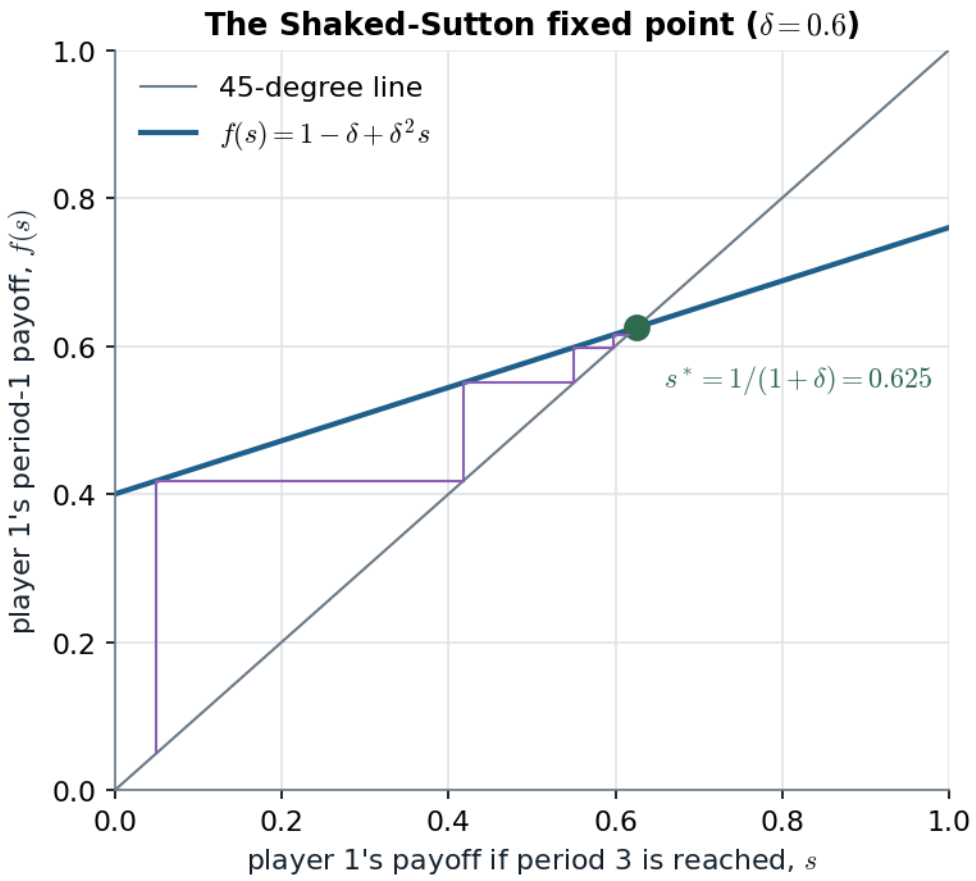


Figure — f has slope $\delta^2 < 1$, so it crosses the diagonal exactly once, at $s^* = 1/(1+\delta)$; iterating f from any start converges there. Source: Gibbons p. 71.

The slope $\delta^2 < 1$ is the whole proof of uniqueness in one picture: a contraction has one fixed point. Iterating f from $s = 1$ reproduces the odd-T shares of note 06 (1, $1 - \delta + \delta^2$, …).

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3 · C — verifying the strategies by one-stage deviations

Use L04-03-one-stage-deviation: with discounting it applies to this infinite game. Check player 2 as responder in round 1, offered $1-x$:

- accept: $1-x$;
- reject, then follow the strategies: she proposes in round 2 and keeps $1-\frac{\delta}{1+\delta}=\frac{1}{1+\delta}$, worth $\frac{\delta}{1+\delta}$ today.

Accepting is optimal iff $1-x\geq\frac{\delta}{1+\delta}$, i.e. $x\leq\frac{1}{1+\delta}$: exactly the rule. ✓

Player 1 as proposer: any $x\leq\frac{1}{1+\delta}$ is accepted, so the largest, $\frac{1}{1+\delta}$, is best; a larger x is rejected and yields at most $\delta\cdot\frac{\delta}{1+\delta}<\frac{1}{1+\delta}$. ✓ Even rounds are the mirror image.

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4 · C — different discount factors: patience is power

Let player i discount at δ_i. Stationary shares: player 1 proposing keeps x; player 2 proposing keeps y (her own share). Each proposer gives the responder exactly the responder's discounted proposer share:

$$x=1-\delta_2 y,\qquad y=1-\delta_1 x.$$

Substitute: $x=1-\delta_2(1-\delta_1 x)\Rightarrow x(1-\delta_1\delta_2)=1-\delta_2$:

$$\boxed{x^*=\frac{1-\delta_2}{1-\delta_1\delta_2},\qquad 1-x^*=\frac{\delta_2(1-\delta_1)}{1-\delta_1\delta_2}}$$

(the case Tadelis sets as Exercise 11.6, p. 237, and MWG 9.B.12, p. 302; derived also in `Aula/aula-02-notes.md` §5.2). With $\delta_1=\delta_2=\delta$, $x^*=\frac{1-\delta}{1-\delta^2}=\frac{1}{1+\delta}$. ✓

Signs (differentiate).

$$\frac{\partial x^*}{\partial\delta_1}=\frac{(1-\delta_2)\,\delta_2}{(1-\delta_1\delta_2)^2}>0,\qquad \frac{\partial x^*}{\partial\delta_2}=\frac{-(1-\delta_1\delta_2)+(1-\delta_2)\delta_1}{(1-\delta_1\delta_2)^2}=\frac{-(1-\delta_1)}{(1-\delta_1\delta_2)^2}<0.$$

> **Your own patience raises your share; your opponent's patience lowers it.** A patient responder is expensive to buy off because waiting costs her little.

Worked ($\delta_1=0.9$, $\delta_2=0.7$, as in `check_part2.py`): $x^*=\frac{0.3}{1-0.63}=\frac{0.3}{0.37}=0.811$, against 0.93 with three rounds (note 06 §6): the deadline was worth 0.12 to player 1, the last mover.

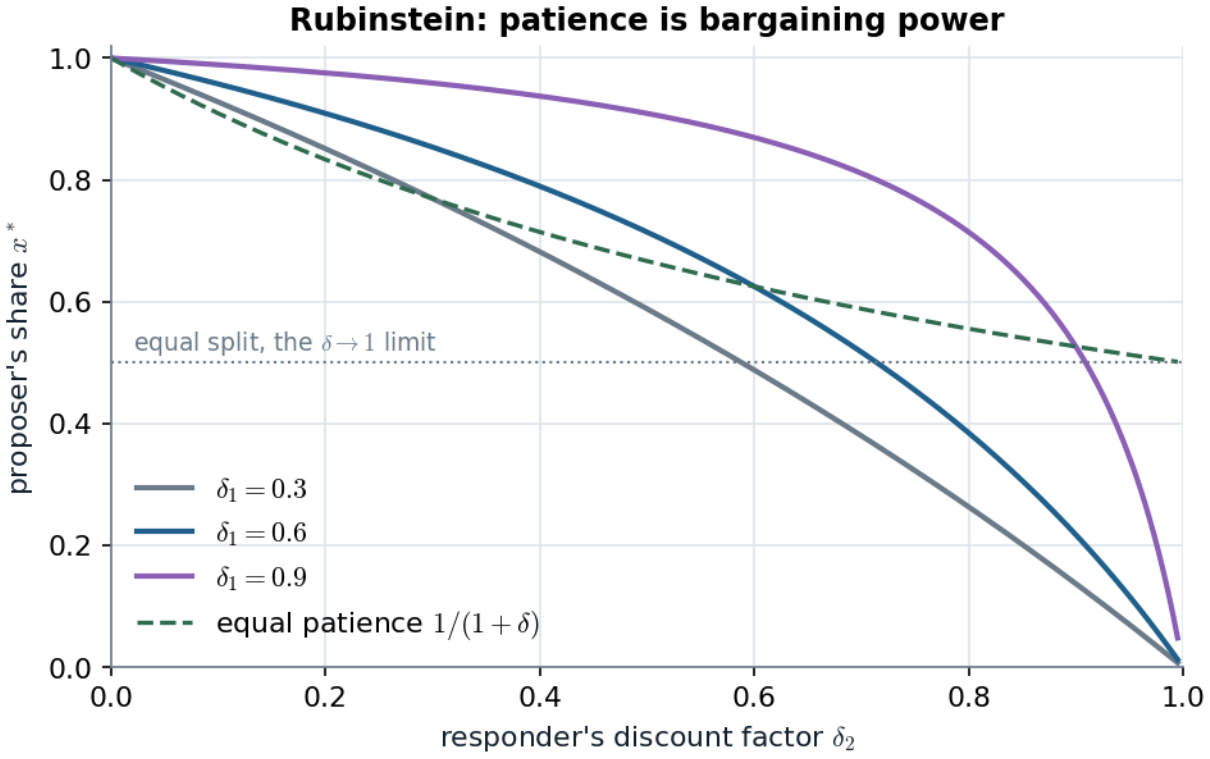


Figure — the proposer's share falls as the responder grows more patient and rises with the proposer's own patience; the dashed curve is the equal-patience case. Source: Tadelis eqs. (11.4)–(11.5), p. 229, extended to $\delta_1 \neq \delta_2$.

Each solid curve starts at 1 (an impatient responder accepts anything) and falls to 0 as $\delta_2 \to 1$ (a responder who does not mind waiting can hold out for everything).

> **Interactive companion.** The Patience Ledger (`companions/companion-rubinstein-bargaining.html`) — tick "No deadline" and move δ_1 and δ_2 separately; the read-out names who is more patient and the dashed line in the left panel is this formula. "Patient pair, no deadline" sets $\delta = 0.99$, a nearly equal split. At $\delta_1 = \delta_2 = 1$ the page reports the share as indeterminate rather than computing $0/0$.

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5 · C — frequent offers and the Nash bargaining solution

Write $\delta_i = e^{-r_i \Delta}$, where r_i is i's discount rate and Δ the time between offers. As $\Delta \to 0$ (by l'Hôpital, or $1 - e^{-z} \approx z$):

$$x^* = \frac{1 - e^{-r_2 \Delta}}{1 - e^{-(r_1 + r_2)\Delta}} \longrightarrow \frac{r_2}{r_1 + r_2}.$$

That is the **generalized Nash bargaining solution** with threat points 0 and weight $\beta_1 = r_2/(r_1 + r_2)$ for player 1 (Aula 2 notes §5.1–5.2; Tadelis's Remark on Nash's axiomatic approach, p. 229). Equal rates give $\frac{1}{2}$, matching (11.3), and the first-mover advantage vanishes because a round lasts no time.

> **The axiomatic weights of the Nash solution have a strategic foundation: relative impatience.** `Listas/lista-aulas-01-02.pdf` Block 3, Question 5b examines exactly this

limit.

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6 · G — MWG's proof of uniqueness, written with bounds (Example 9.AA.2, pp. 297–298)

Pie v. Let $\bar{v}_1$ and $\underline{v}_1$ be player 1's highest and lowest SPE payoffs.

1. **Lower bound.** Player 2 can get at most $\bar{v}_1$ as the round-2 proposer, worth $\delta\bar{v}_1$. Offering her slightly more is surely accepted, so $\underline{v}_1 \geq v - \delta\bar{v}_1$.
2. **Upper bound.** Player 2 can get at least $\underline{v}_1$ as round-2 proposer and rejects any round-1 offer below $\delta\underline{v}_1$; and a rejected offer leaves player 1 at most $\delta v - \delta\underline{v}_1 < v - \delta\underline{v}_1$. So $\bar{v}_1 \leq v - \delta\underline{v}_1$.
3. **Squeeze.** $\bar{v}_1 \leq v - \delta\underline{v}_1 \leq v - \delta(v - \delta\bar{v}_1)$, so $\bar{v}_1(1-\delta^2) \leq v(1-\delta)$ and $\bar{v}_1 \leq \frac{v}{1+\delta}$. Then $\underline{v}_1 \geq v - \delta\bar{v}_1 \geq v - \frac{\delta v}{1+\delta} = \frac{v}{1+\delta}$.

$$\frac{v}{1+\delta} \leq \underline{v}_1 \leq \bar{v}_1 \leq \frac{v}{1+\delta} \quad \Longrightarrow \quad v_1^* = \frac{v}{1+\delta},\ v_2^* = \frac{\delta v}{1+\delta}.$$

MWG add a warning (p. 298): the result is **sensitive to the protocol and to preferences**; with a fixed cost per round instead of discounting (Exercises 9.B.7, 9.B.13) the infinite game can have many SPE.

Links

- Previous / next: L04-06-finite-alternating-offers · L04-08-legislative-bargaining.
- Earlier project material: `Aula/aula-02-notes.md` §5 (Nash solution, alternating offers); `Aula/notes-aulas-01-02/main.pdf` part 2; `Listas/lista-aulas-01-02.pdf` Block 3 (Questions 5–6) and its check `Listas/lista-aulas-01-02-check.py`; quiz `Simulados/quiz-aula02-3-nash-offers-2026-10-13.md`.
- Python: L04-12-python §3. Pitfalls: L04-11-pitfalls items 9–12.
- Rules pack: rules/02_barganha.
- **Exercises (number and page only):** Tadelis 11.6 (p. 237); Gibbons Problem 2.3 (p. 131); MWG 9.B.12 (pp. 302–303); Muñoz-Garcia Ch. 6 Ex. 8 (p. 303).

11.4 · Legislative bargaining: majority rule, random proposers, and amendments

> **Claim.** Replace "both must agree" by "a majority must agree" and "alternate" by "a random legislator proposes" (Baron and Ferejohn, 1989). In the symmetric stationary equilibrium the proposer buys the **cheapest** majority, paying each partner the discounted continuation value δ/N. Under a **closed rule** (no amendments) that is worth $1-\delta\frac{N-1}{2N}$ to the proposer; an **open rule** destroys targeting and cuts the proposer to $\frac{1}{1+2\delta}$ when $N=3$, unless she gambles on a minimal coalition, which pays when $\delta>\sqrt{3}-1$.

Prev: L04-07-rubinstein-infinite-horizon · Next: L04-09-bargaining-private-information · Map: L04-00-map

> **Scope note.** §11.4 lies inside the study guide's page range (pp. 220–237). Whether the lecture covers it is unknown until the slides arrive; the section is tagged C on that basis and should be rechecked at reconcile.

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1 · C — the two departures (pp. 229–230)

1. N players, N odd; $\frac{N+1}{2}$ votes pass a proposal.
2. Each period **every player is equally likely (prob. $1/N$) to be the proposer**. No fixed order, so no last mover.

With three or more players the game has many SPE (Herrero, 1985; footnote 4, p. 230). Baron and Ferejohn select **symmetric stationary** equilibria: a proposer makes the same proposal whenever called on, and responders vote on the current proposal against their expected continuation. Then the game "starts over" after each rejection, and every player's expected value at the start of a period is the same number v (pp. 230–231).

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2 · C — closed rule (§11.4.1, pp. 230–231)

Responders. Rejecting is worth δv, so responder i accepts iff $x_i \geq \delta v$.

Proposer. She needs $\frac{N+1}{2}$ votes, her own plus $\frac{N-1}{2}$ others. She picks $\frac{N-1}{2}$ partners at random, pays each exactly δv, gives the rest 0, and keeps

$$k = 1 - \frac{N-1}{2}\,\delta v. \tag{11.6}$$

Value of the game. With probability $\frac{1}{N}$ a player proposes (k); with probability $\frac{N-1}{N}$ she responds, and then is chosen as a partner with probability $\frac{1}{2}$ (getting δv), else 0:

$$v = \frac{k}{N} + \frac{N-1}{2N}\,\delta v. \tag{11.7}$$

Solve. Substitute (11.6): $v = \frac{1}{N} - \frac{N-1}{2N}\delta v + \frac{N-1}{2N}\delta v = \frac{1}{N}$. Then

$$\boxed{v = \frac{1}{N}, \qquad k(N) = 1 - \delta\,\frac{N-1}{2N}} \tag{11.8}$$

$v = 1/N$ is forced by symmetry and immediate agreement; the interesting object is k.

Worked.

	$\delta = 0.9$	$\delta \to 1$
$N = 3$: $k = 1 - \delta/3$	$1 - 0.3 = \mathbf{0.70}$; each partner $\delta/3 = 0.30$	$2/3$
$N = 5$: $k = 1 - \delta \cdot 4/10$	$1 - 0.36 = \mathbf{0.64}$; two partners at 0.18	0.6
$N \to \infty$: $k \to 1 - \delta/2 = \frac{2-\delta}{2}$	0.55	$1/2$

Two readings (pp. 231–232).

1. As in Rubinstein, k **falls with** δ: patient responders must be paid more.
2. As N grows, each partner costs less (δ/N) but more partners are needed; the second force wins and k falls. Yet relative to a partner the proposer's share $k/(\delta/N)$ **rises without bound**: more players means more ways to form a coalition.

---

3 · C — open rule, guaranteed success (§11.4.2.1, pp. 232–233), $N = 3$

After a proposal, one of the other two is drawn as **amender**. She can second it (then a majority vote) or offer an amendment, which is voted against it and becomes the new baseline next period.

Guaranteed success. The proposer keeps k and offers each of the others $\frac{1-k}{2}$, so whoever is drawn seconds it. The amender's alternative is to propose herself next period, worth δk by symmetry. The proposer pays exactly that:

$$\frac{1-k}{2} = \delta k \iff 1 - k = 2\delta k \iff \boxed{k = \frac{1}{1+2\delta}} \tag{11.9}$$

Closed beats open (footnote 7, p. 233).

$$\left(1 - \frac{\delta}{3}\right) - \frac{1}{1+2\delta} = \frac{(3-\delta)(1+2\delta) - 3}{3(1+2\delta)} = \frac{5\delta - 2\delta^2}{3(1+2\delta)} = \frac{\delta(5-2\delta)}{3(1+2\delta)} > 0.$$

Under a closed rule the proposer can play some legislators against others; under an open rule a shortchanged legislator amends at her expense. As $\delta \to 1$, $k \to \frac{1}{3}$: an equal split, against $\frac{2}{3}$ under the closed rule (p. 233).

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4 · C — open rule, risky success (§11.4.2.2, pp. 233–235)

The proposer offers $1 - k$ to **one** partner and 0 to the third. If the partner is drawn as amender she seconds; if the excluded player is drawn she amends to (k for herself, $1 - k$ to the same partner, 0 to the original proposer) and wins ("selective inclusion"). Let $v(k)$ be the proposer's expected payoff and $v(0)$ the excluded player's.

$$v(k) = \tfrac{1}{2}k + \tfrac{1}{2}\,\delta v(0) \tag{11.10}$$

$$v(0) = \tfrac{1}{2}\,\delta v(k) \tag{11.11}$$

$$1 - k = \delta v(k) \tag{11.12}$$

(11.10): with prob. $\frac{1}{2}$ the partner seconds and the proposer gets k; otherwise she is the excluded one next period. (11.11): with prob. $\frac{1}{2}$ the excluded player is drawn and becomes proposer next period. (11.12): the partner could reject and stand in the proposer's shoes next period.

Solve. Put (11.11) in (11.10): $v(k) = \frac{k}{2} + \frac{\delta^2}{4}v(k)$, so $v(k) = \frac{2k}{4-\delta^2}$. Then (11.12): $1 - k = \frac{2\delta k}{4-\delta^2}$, so $k\,(4 - \delta^2 + 2\delta) = 4 - \delta^2$:

$$\boxed{k = \frac{4-\delta^2}{4+2\delta-\delta^2}, \qquad v(k) = \frac{2}{4+2\delta-\delta^2}} \tag{11.13}$$

Closed still beats it (footnote 8, p. 235): $(1 - \frac{\delta}{3}) - \frac{2}{4+2\delta-\delta^2} = \frac{6+2\delta-5\delta^2+\delta^3}{12+6\delta-3\delta^2} > 0$ (numerator $(3-\delta)(4+2\delta-\delta^2) - 6$; at $\delta = 1$ it is 4).

Risky against guaranteed (p. 235).

$$\frac{2}{4+2\delta-\delta^2} > \frac{1}{1+2\delta} \iff 2 + 4\delta > 4 + 2\delta - \delta^2 \iff \delta^2 + 2\delta - 2 > 0 \iff \boxed{\delta > \sqrt{3} - 1 \approx 0.732}$$

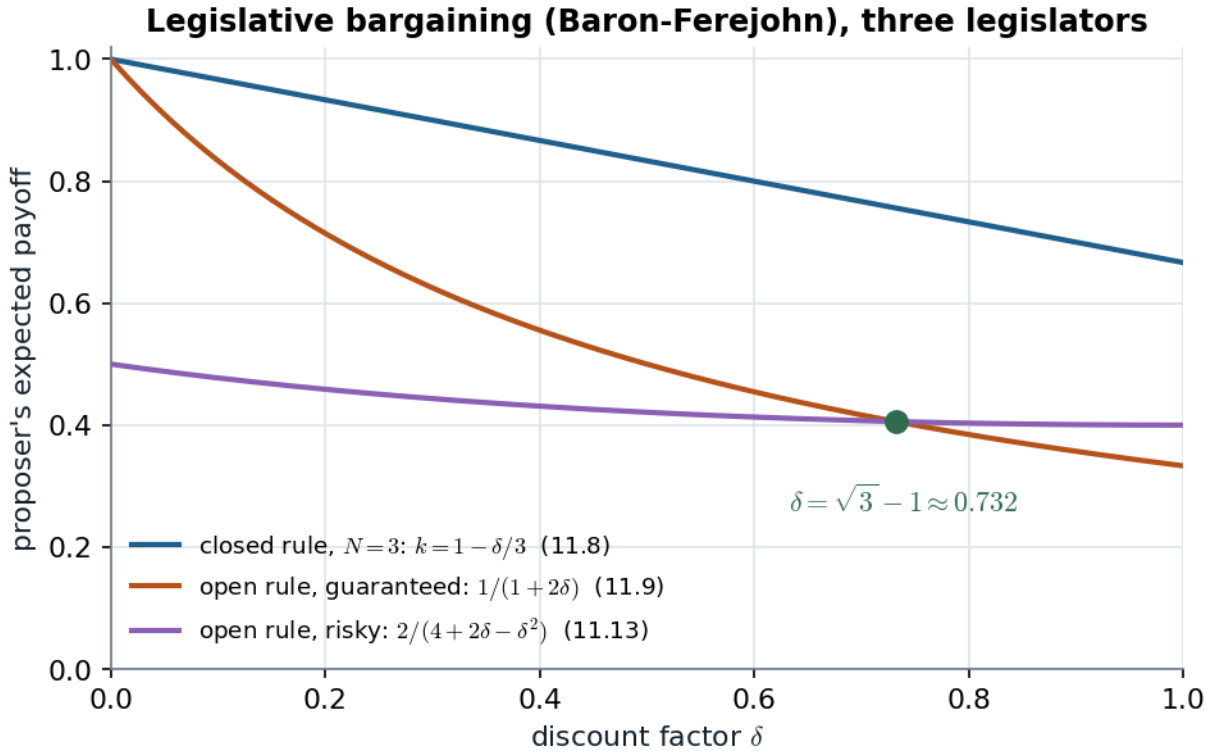


Figure — (11.8) with $N = 3$, (11.9) and (11.13). The closed rule dominates both open-rule payoffs; the risky strategy beats the guaranteed one to the right of $\sqrt{3} - 1$. Source: Tadelis pp. 231–235.

Worked, $\delta = 0.9$: closed 0.700; guaranteed $1/2.8 = 0.357$; risky $k = 3.19/4.99 = 0.639$ with expected payoff $2/4.99 = 0.401 > 0.357$. Patient legislators make buying off **both** potential amenders too expensive, so the proposer caters to one and accepts a 50% chance of failure and **delay**, which, unlike in Rubinstein's game, now happens in equilibrium (p. 235).

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5 · U — what the chapter summary asks you to keep (§11.5, pp. 235–236)

Summary point	Where
The discount factor drives the split; more patience, more equal shares	notes 06–07, §2 here
Protocol and agreement rules matter a lot	closed vs open rule, §3–§4
Finite alternating games: last-mover advantage; discounting: first-mover advantage	note 06
Infinite horizon or random proposers: **no last-mover advantage**, only the first-mover one	note 07, §2 here

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Links

- Previous / next: L04-07-rubinstein-infinite-horizon · L04-09-bargaining-private-information.
- Python: L04-12-python §4 (closed and open rule, the $\sqrt{3} - 1$ threshold).
- Pitfalls: L04-11-pitfalls item 13.
- **Exercises (number and page only):** Tadelis 11.7 (p. 237); Muñoz-Garcia Ch. 6 Ex. 8 (p. 303).

M&R · Bargaining over a sale: where the complete-information prediction stops

> **Claim.** Notes 05–08 predict **immediate, efficient agreement** whenever values are common knowledge (Proposition 11.3). M&R's two-type example shows the prediction depends on that assumption: once each side privately knows its own value, **no price rule** can make both report truthfully *and* trade whenever trade is efficient. The seller needs $p \geq \$1.75$, the buyer $p \leq \$1.25$. Bargaining power then decides more than the split; it decides whether value is created.

Prev: L04-08-legislative-bargaining · Next: L04-10-formulas · Map: L04-00-map

> **Layer.** U (application), from the study guide's "Applications" line. The general theory behind it, mechanism design and the revelation principle, is **out of scope** (`CLAUDE.md`); it is named here only so the result can be placed. Incomplete information proper is Lecture 7.

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1 · U — the organizational point (M&R p. 140)

With no wealth effects the **value-creating** part of an agreement is determinate (the value maximization principle, `Aula/aula-02-notes.md` §3); bargaining strength, patience and guile decide only the **transfers**. That is exactly what Rubinstein's model delivers: the pie is never wasted and δ sets the split. M&R's caveat: efficiency is defined relative to what is **feasible**, and what is feasible depends on information.

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2 · Worked — the example (pp. 140–142)

Values. Seller's value 0 (prob. 0.8) or \$2 (prob. 0.2); buyer's value \$1 (prob. 0.2) or \$3 (prob. 0.8). Each knows only its own value.

Efficient outcome (Table 5.2). Trade unless buyer \$1 and seller \$2, which has probability $0.2 \times 0.2 = 0.04$. **Trade is efficient 96% of the time.**

A mediator's prices (Table 5.3). Look for prices that make truthful reports optimal:

Reports	Price	Why
seller 2, buyer 3	\$2	a seller claiming 2 must get at least 2; more would only reward the claim
seller 0, buyer 1	\$1	a buyer claiming 1 cannot be asked more than 1
seller 2, buyer 1	no trade	inefficient
seller 0, buyer 3	p	to be found

Seller with value 0 (incentive compatibility).

- Honest: she trades at \$1 with a low buyer (prob. 0.2) and at p with a high one (prob. 0.8): $0.2(1) + 0.8p$.
- Claims 2: no trade with a low buyer; \$2 with a high one: $0.2(0) + 0.8(2) = 1.60$.

$$0.2 + 0.8p \geq 1.60 \iff p \geq \frac{1.40}{0.8} = \$\mathbf{1.75}.$$

Buyer with value 3.

- Honest: pays \$2 against a high-value seller (prob. 0.2), p against a low one (prob. 0.8): $0.2(3-2) + 0.8(3-p) = 2.60 - 0.8p$.
- Claims 1: trades only with the low seller, at \$1: $0.8(3-1) = 1.60$.

$$2.60 - 0.8p \geq 1.60 \iff p \leq \frac{1.00}{0.8} = \$\mathbf{1.25}.$$

$p \geq 1.75$ **and** $p \leq 1.25$ **cannot both hold. There is no price schedule under which both report truthfully and trade happens exactly when it is efficient (p. 142).** Footnote 8 adds that allowing misreporting does not help; that is where the revelation principle enters, out of scope here.

The same numbers are verified in `Aula/notes-aulas-01-02/check_part2.py` (block "M&R Table 5.2/5.3"), together with the reversed-probability case where efficient trade *is* feasible.

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3 · U — the bridge back to the game theory

	Complete information (notes 05–08)	Private values (this note)
Delay	never on the equilibrium path (Prop. 11.3)	can be a costly way to signal strength
Efficiency	always reached	can be unattainable by any rule
What bargaining power decides	the split	the split **and** whether trade happens

Gibbons anticipates the same contrast when he opens §2.1.D (p. 68): in Rubinstein's model "settlement occurs immediately", whereas in sequential bargaining under asymmetric information strikes occur with positive probability (Sobel and Takahashi, 1983; his §4.3.B), which belongs with Lecture 8's perfect Bayesian equilibrium.

> **For the organization designer: the complete-information model tells you how surplus is split once the parties can see it; the private-information example tells you why so much effort goes into making it visible** (audits, standard costs, third-party appraisals).

Links

- Previous / next: L04-08-legislative-bargaining · L04-10-formulas.
- Earlier material: `Aula/notes-aulas-01-02/main.pdf` part 2; `Listas/pratica-aula-01-02.md` block B3 (items 21–23, 30).
- Rules packs: rules/02_barganha, rules/07_selecao_adversa.
- Python: L04-12-python §5.

Lecture 4 · Formulas — every definition, result and number, the study guide mapped

Prev: L04-09-bargaining-private-information · Next: L04-11-pitfalls · Map: L04-00-map

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0 · The exam specification, mapped (estimated)

No slides yet, so the specification is the study guide's Lecture 4 block (`Map/study-guide.md`), item by item. Reconcile with the slides' learning goals when the Lecture 4 slides arrive.

#	Study-guide item	What to be able to do	Note
1	Tadelis Ch. 9 §9.1–9.3 (pp. 175–180)	define a multistage game; write $v_i = \sum_t \delta^{t-1} v_{it}$; write strategies as history-contingent plans and count them	L04-01-multistage-games
2	Tadelis §9.4 (pp. 180–184)	state Props. 9.1–9.3; build a carrot-and-stick SPE; derive δ^*	L04-02-carrot-and-stick
3	Tadelis §9.5 (pp. 184–186)	state and apply the one-stage deviation principle	L04-03-one-stage-deviation
4	Gibbons §2.2 (pp. 71–82)	solve two-stage games by "stage-2 NE first": bank runs, tariffs, tournaments	L04-01-multistage-games §5; L04-04-two-stage-applications
5	Tadelis §11.1 (pp. 222–224)	ultimatum game: NE vs SPE (Props. 11.1–11.2)	L04-05-ultimatum-game
6	Tadelis §11.2 (pp. 224–227); Gibbons §2.1.D (pp. 68–70); MWG Ex. 9.AA.1	finite alternating offers by backward induction; (11.1); Prop. 11.3; limits and their order	L04-06-finite-alternating-offers
7	Tadelis §11.3 (pp. 228–229); Gibbons pp. 70–71; MWG Ex. 9.AA.2	Rubinstein by stationarity; asymmetric patience; $\Delta \to 0$ and Nash	L04-07-rubinstein-infinite-horizon
8	Tadelis §11.4 (pp. 229–235)	closed and open rule legislative bargaining	L04-08-legislative-bargaining
9	M&R bargaining over a sale (p. 140)	why private values can make efficient trade infeasible	L04-09-bargaining-private-information
10	`Aula/aula-02-notes.md` §5; Listas Block 3; `rules/02_barganha.md`	link Rubinstein to Nash bargaining and to Coase	L04-07-rubinstein-infinite-horizon §5

1 · Every definition and proposition

1.1 · Multistage games (Tadelis Ch. 9)

Item	Statement	Note
Multistage game (p. 176)	finite sequence of simultaneous stage-games, same players, each outcome observed before the next stage	01 §2
Payoffs (p. 177)	$v_i = \sum_{t=1}^{T} \delta^{t-1} v_{it}$	01 §3
Pure strategy (p. 179)	$s_i = \{s_{i1}, s_{i2}(h_1), \dots, s_{iT}(h_{T-1})\}$, one action per history	01 §4
Prop. 9.1 (p. 180)	any sequence of stage NE, played unconditionally, is an SPE path	02 §1
Prop. 9.2 (p. 181)	in any NE, play in the last stage is a stage NE	02 §2
Prop. 9.3 (p. 181)	every stage has a unique NE ⇒ unique SPE	02 §2
Requirements (p. 183)	(1) two distinct stage-2 equilibria, a carrot and a stick; (2) δ large enough	02 §4
Threshold	$\delta^* = g/(c-s)$: one-shot gain over carrot minus stick	02 §4
Def. 9.1 (p. 185)	one-stage unimprovable: no single information set where changing the action alone pays	03 §2
Thm. 9.1 (p. 185)	one-stage unimprovable ⇒ optimal (finite trees)	03 §2

1.2 · Two-stage applications (Gibbons §2.2)

Item	Statement	Note
Subgame-perfect outcome (p. 73)	$(a_1^*, a_2^*, a_3^*(a_1^*, a_2^*), a_4^*(a_1^*, a_2^*))$	01 §5
Bank runs (p. 75)	two SPE outcomes: run (r, r) and no run (R, R)	04 §1
Stage-2 Cournot with tariffs (2.2.3)	$h_i^* = \frac{a-c+t_i}{3}$, $e_i^* = \frac{a-c-2t_j}{3}$	04 §2.1
Dominant tariff (p. 78)	$t_i^* = \frac{a-c}{3}$; $h^* = \frac{4(a-c)}{9}$, $e^* = \frac{a-c}{9}$; joint optimum $t = 0$	04 §2.2
Tournament FOC (2.2.6)	$(w_H - w_L) \int f^2 = g'(e^*)$; normal noise $\int f^2 = \frac{1}{2\sigma\sqrt{\pi}}$	04 §3
Optimal prize (p. 82)	$g'(e^*) = 1$ and $(w_H - w_L) \int f^2 = 1$; participation binds (2.2.8)	04 §3

1.3 · Bargaining (Tadelis Ch. 11, Gibbons §2.1.D, MWG 9.AA)

Item	Statement	Note
Prop. 11.1 (p. 223)	$T=1$: every $(x^*, 1-x^*)$ is a NE	05 §2
Prop. 11.2 (p. 223)	$T=1$: unique SPE, offer $x=1$, accept any $x \le 1$	05 §3
Two rounds (p. 225)	$(v_1, v_2) = (1-\delta, \delta)$	06 §2
(11.1) (p. 226)	T odd: $v_1 = \frac{1+\delta^T}{1+\delta}$, $v_2 = \frac{\delta-\delta^T}{1+\delta}$; all T: $x_1 = \frac{1-(-\delta)^T}{1+\delta}$	06 §3
Prop. 11.3 (p. 226)	every SPE agrees in round 1	06 §4
(11.2)–**(11.3)** (p. 227)	$\lim_T v_1 = \frac{1}{1+\delta}$; then $\lim_{\delta\to 1} = \frac{1}{2}$; reverse order keeps $v_1 = 1$	06 §5
Gibbons three periods (p. 70)	$s_1^* = 1-\delta(1-\delta s)$	06 §6
Asymmetric, 3 rounds	$s_1 = 1-\delta_2(1-\delta_1)$	06 §6
(11.4)–**(11.5)** (p. 229)	$\underline{v} = 1-\delta\bar{v}$, $\bar{v} = 1-\delta\underline{v} \Rightarrow v = \frac{1}{1+\delta}$	07 §2
Fixed point (Gibbons p. 71)	$f(s) = 1-\delta+\delta^2 s$, $s^* = \frac{1}{1+\delta}$	07 §2.1
Asymmetric, infinite	$x^* = \frac{1-\delta_2}{1-\delta_1\delta_2}$; $\partial x^*/\partial\delta_1 > 0$, $\partial x^*/\partial\delta_2 < 0$	07 §4
Frequent offers	$\delta_i = e^{-r_i\Delta}$, $\Delta\to 0$: $x^* \to \frac{r_2}{r_1+r_2}$ = Nash weight	07 §5
MWG 9.AA.2 (pp. 297–298)	squeeze $\frac{v}{1+\delta} \le \underline{v}_1 \le \bar{v}_1 \le \frac{v}{1+\delta}$	07 §6
(11.6)–**(11.8)** (pp. 230–231)	closed rule: $v = \frac{1}{N}$, $k = 1-\delta\frac{N-1}{2N}$	08 §2
(11.9) (p. 233)	open rule, guaranteed: $k = \frac{1}{1+2\delta}$	08 §3
(11.10)–**(11.13)** (p. 234)	open rule, risky: $k = \frac{4-\delta^2}{4+2\delta-\delta^2}$, $v(k) = \frac{2}{4+2\delta-\delta^2}$	08 §4
Risky vs guaranteed (p. 235)	risky better iff $\delta > \sqrt{3}-1$	08 §4

1.4 · Private information (M&R)

Item	Statement	Note
Incentive constraints (pp. 141–142)	seller needs $p \ge 1.75$, buyer needs $p \le 1.25$: no truthful efficient rule	09 §2

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2 · Every number and worked case

Case	Numbers	Source	Note
Prisoner-Revenge strategies	$2^5 = 32$ each, 1024 profiles	p. 179	01 §4
Revenge mixed NE	$q = \frac{1}{2}$, value -2	p. 177	01 §2.1

Case	Numbers	Source	Note
Discounted payoff example	(F, m) then (L, g): $v_1 = 5 - 4\delta$, $v_2 = -1 - \delta$	p. 177	01 §3
Support (M, m)	$4 \geq 5 - 3\delta \iff \delta \geq \frac{1}{3}$	p. 183	02 §3
Support (F, m)	$-1 \geq 1 - 3\delta \iff \delta \geq \frac{2}{3}$	p. 184	02 §5
Reduced stage 1 at $\delta = 0.5$	$(M, m) = (4, 4)$, $(F, m) = (3.5, -2.5)$, $(F, f) = (-0.5, -0.5)$	from p. 182	02 §3.1
Bank run (illustrative)	$D = 100, r = 75, R = 150$: NE $(75, 75)$ and $(150, 150)$	pp. 73–75	04 §1
Tariff game	$72, 81, 56, 65$ in units $(a - c)^2/162$	p. 78	04 §2.2
Tournament (illustrative)	$\sigma = 1$: spread $2\sqrt{\pi} \approx 3.54$	pp. 81–82	04 §3
Ultimatum experiments	offers 25–50%, rejections below 20–25%	pp. 223–224	05 §4
Finite shares, $\delta = 0.6$	$T = 1..6$: 1, 0.4, 0.76, 0.544, 0.674, 0.596; limit 0.625	(11.1)	06 §3.1
Asymmetric $\delta_1 = 0.9, \delta_2 = 0.7$	3 rounds 0.93; infinite $0.3/0.37 = 0.811$	`check_part2.py`	06 §6, 07 §4
Rubinstein, equal δ	$0.6 \to 0.625$; $0.9 \to 0.526$	(11.4)–(11.5)	07 §2
Closed rule, $\delta = 0.9$	$N = 3$: 0.70; $N = 5$: 0.64; $N \to \infty$: 0.55	(11.8)	08 §2
Open rule, $\delta = 0.9$	guaranteed 0.357; risky $k = 0.639$, $v(k) = 0.401$	(11.9), (11.13)	08 §4
Threshold	$\sqrt{3} - 1 \approx 0.732$	p. 235	08 §4
M&R sale	efficient trade 96%; $p \geq 1.75$ vs $p \leq 1.25$	pp. 140–142	09 §2

Lecture 4 · Pitfalls and an original True/False self-test

Prev: L04-10-formulas · Next: L04-12-python · Map: L04-00-map

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Part I — the confusions this material produces

1 · "An SPE plays a stage Nash equilibrium in every stage"

Only the last stage must be a stage NE (Proposition 9.2). Earlier stages can be anything the continuation can enforce. *Counterexample:* (M, m) in stage 1 of the Prisoner-Revenge Game, an SPE path for $\delta \geq \frac{1}{3}$, although (M, m) is not a NE of the Prisoner's Dilemma. → L04-02-carrot-and-stick §3.

2 · "Two equilibria in stage 2 are enough"

They are necessary, not sufficient: the gap must also beat the one-shot gain after discounting, $g \leq \delta(c-s)$. *Counterexample:* at $\delta = 0.2$ only (F, f) survives in stage 1. And with a **unique** stage-2 NE nothing can be enforced at all (Proposition 9.3). → note 02 §2, §4.

3 · "Punish with any threat that hurts"

A punishment must be a **Nash equilibrium of the subgame** where it is carried out. "I play gang while you play loner" is not one (-1 for the gang member, but the loner would switch to G for $-3 > -4$). Only the switch from (L, l) to (G, g) is credible. → note 02 §4.

4 · Testing a deviation with the wrong continuation

A one-stage deviation changes the action at **one** information set; afterwards the deviator follows **his own** strategy, which may itself prescribe the punishment. *Example:* the stage-1 fink is followed by G, so the deviation payoff is $5 - 3\delta$, not $5 + 0$. → L04-03-one-stage-deviation §3.

5 · Two-stage games: substituting a non-equilibrium, or reading dominance as efficiency

Stage 2 is replaced by its **Nash equilibrium**, never by the jointly best outcome. And a dominant tariff is not a good tariff: (t^*, t^*) gives 65/162 each against 72/162 under free trade. Nor does the bank-run model predict runs; it shows they *can* be an equilibrium. → L04-04-two-stage-applications.

6 · Nash vs subgame perfect in the ultimatum game

"Any split is an equilibrium" is true **for Nash** (Proposition 11.1) and false for SPE (Proposition 11.2). The Nash strategies rely on a responder who would reject a positive offer, a threat never tested on the path. → L04-05-ultimatum-game.

7 · "The experiments refute the theory"

They refute *money = utility*. With a payoff that falls with the proposer's advantage, rejecting a low offer is sequentially rational, and the SPE changes accordingly. → note 05 §4.

8 · Mixing the two advantages

The **last** mover's advantage comes from the deadline; the **first** mover's from discounting. With $\delta = 1$ only the **last** word matters ($x_1 = 1$ for odd T, 0 for even). With $T \to \infty$ only the **first** move matters, and it is worth $\frac{1}{1+\delta}$. → L04-06-finite-alternating-offers §5.

9 · Offering the responder the discounted *pie* instead of her discounted *continuation*

The responder gets $\delta\times$ what she would obtain **as next round's proposer**, not $\delta \times 1$. With two rounds those coincide (δ), with three they do not ($\delta(1-\delta)$ to player 2). → note 06 §3; L04-07-rubinstein-infinite-horizon §3.

10 · The order of limits

$\lim_{\delta\to 1}\lim_{T\to\infty} v_1 = \frac{1}{2}$ but $\lim_{T\to\infty}\lim_{\delta\to 1} v_1 = 1$ (odd T). "Patient players split equally" needs the horizon to go first (footnote 2, p. 227).

11 · Swapping the discount factors

$x^* = \frac{1-\delta_2}{1-\delta_1\delta_2}$ has the **responder's** factor in the numerator. Your own patience raises your share, the opponent's lowers it. The swapped formula $\frac{1-\delta_1}{1-\delta_1\delta_2}$ is a standard wrong answer (it is distractor D of Listas Block 3, Question 5a). → note 07 §4.

12 · "The infinite game is the limit of the finite games"

Here the answers coincide, but that is a **theorem**, proved by stationarity, not a definition. In repeated games (Lecture 5) the infinite game supports cooperation that every finite version rules out (MWG p. 298). → note 07 §1.

13 · Legislative bargaining: $v = 1/N$ is not the proposer's share

$v = 1/N$ is the *ex ante* value of the game; the proposer keeps $k = 1 - \delta\frac{N-1}{2N}$, more than $1/N$. And an open rule hurts the proposer: $\frac{1}{1+2\delta} < 1 - \frac{\delta}{3}$. → L04-08-legislative-bargaining.

Part II — True or False (original, 20 items)

Scoring as in the course style: $\max\{0,\ \text{correct} - \text{wrong}/2\}$ per item value; blanks do not count.

1. In a finite multistage game in which every stage-game has a unique Nash equilibrium, the only subgame-perfect path plays that equilibrium in every stage. **T.** Proposition 9.3: by backward induction the last stage is fixed after every history, so the second-to-last stage cannot be linked to the future, and so on.

2. In the Prisoner-Revenge Game, (M, m) in stage 1 can be part of an SPE path when $\delta = 0.4$. **T.** The condition is $4 \geq 5 - 3\delta$, i.e. $\delta \geq \frac{1}{3}$; 0.4 qualifies ($4 \geq 3.8$).

3. In the same game, (F, m) in stage 1 can be part of an SPE path when $\delta = 0.6$. **F.** Player 2 needs $-1 \geq 1 - 3\delta$, i.e. $\delta \geq \frac{2}{3}$; at 0.6 fink pays $1 - 1.8 = -0.8 > -1$.

4. If stage 2 has two Nash equilibria with the same payoffs for both players, history-dependent strategies can still sustain a non-equilibrium outcome in stage 1. **F.** Carrot minus stick is zero, so the discounted punishment $\delta(c - s) = 0$ cannot offset any positive one-shot gain.

5. A player in the Prisoner-Revenge Game has 32 pure strategies. **T.** One stage-1 action plus one stage-2 action for each of the four stage-1 outcomes: $2^5 = 32$.

6. To check that a profile is subgame perfect in a finite game, it suffices to check that no player gains by changing his action at a single information set while following his strategy elsewhere. **T.** Theorem 9.1, the one-stage deviation principle.

7. In Gibbons's bank-run model, both investors withdrawing at date 1 is the unique subgame-perfect outcome. **F.** After substituting date 2's (R, R), the date-1 game has two pure NE: the run (r, r) and waiting (R, R).

8. In Gibbons's tariff model, each government's equilibrium tariff depends on the other government's tariff. **F.** W_i^* depends on t_j only through the export-profit term, which does not involve t_i; so $t_i^* = (a - c)/3$ is dominant.

9. In the tariff model, both governments would be better off committing to zero tariffs. **T.** Each gets $72/162$ of $(a - c)^2$ at free trade against $65/162$ at the dominant tariffs: a Prisoners' Dilemma.

10. In the ultimatum game, offering the responder 40% of the pie is a Nash equilibrium outcome. **T.** Proposition 11.1: "accept iff offered at least 40%" against "offer 40%" are mutual best responses. It is not subgame perfect.

11. In the ultimatum game's unique SPE, the responder rejects an offer of zero. **F.** If she rejected zero, the proposer would have no best response (no largest $x < 1$); in the SPE she accepts every offer, including $x = 1$ (Proposition 11.2).

12. With two rounds of bargaining and a common $\delta = 0.7$, the first proposer receives 70% of the pie. **F.** She receives $1 - \delta = 30\%$; the responder can wait and take the whole pie in round 2, worth 0.7 now.

13. With $T = 3$ and a common $\delta = 0.5$, player 1 receives 0.75. **T.** $x_1 = 1 - \delta + \delta^2 = 1 - 0.5 + 0.25 = 0.75$, equivalently $\frac{1+0.125}{1.5} = 0.75$.

14. In the finite alternating-offer game with discounting, an agreement may be reached only in the last round if players are very patient. **F.** Proposition 11.3: any later agreement wastes pie that a slightly better round-1 offer could share, so every SPE agrees in round 1, for any $\delta < 1$.

15. In the finite game with $\delta = 1$ and T odd, player 1 receives the whole pie. **T.** With no cost of delay the last proposer's take-it-or-leave-it power flows up the tree: $x_1 = \frac{1+1}{1+1} = 1$.

16. In Rubinstein's infinite-horizon game with common δ, the first proposer receives more than half the pie for every $\delta \in (0, 1)$. **T.** $\frac{1}{1+\delta} > \frac{1}{2} \iff \delta < 1$.

17. In Rubinstein's game with $\delta_1 = 0.8$ and $\delta_2 = 0.5$, player 1's share is $\frac{0.5}{0.6}$. **T.** $\frac{1-\delta_2}{1-\delta_1\delta_2} = \frac{0.5}{1-0.4} = \frac{0.5}{0.6} \approx 0.833$.

18. Raising the responder's discount factor raises the proposer's infinite-horizon share. **F.** $\partial x^*/\partial \delta_2 = -(1 - \delta_1)/(1 - \delta_1\delta_2)^2 < 0$: a patient responder must be paid more.

19. Under Baron–Ferejohn's closed rule with three legislators, the proposer keeps $1 - \delta/3$. **T.** (11.8) with $N = 3$: $k = 1 - \delta\frac{2}{6}$; she buys one partner at $\delta v = \delta/3$.

20. Under the open rule with three legislators, catering to a single partner gives the proposer a higher expected payoff than the guaranteed-success proposal for every $\delta \in (0, 1)$. **F.** Only for $\delta > \sqrt{3} - 1 \approx 0.732$; below that the guaranteed proposal $\frac{1}{1+2\delta}$ is better.

Links

- Formulas: L04-10-formulas. Python checks of every number above: L04-12-python.
- Quiz on the same ground from Aula 2: `Simulados/quiz-aula02-3-nash-offers-2026-10-13.md`.

Lecture 4 · Computation in Python

Every routine the unit needs, each followed by `assert` checks against the books' own numbers. Python only (numpy). The code blocks run top to bottom as one script; paste them into one file or a notebook. **Reuse:** the finite-horizon solver is `proposer_share` from `Aula/notes-aulas-01-02/check_part2.py`, loaded with `ast` so that its module-level checks (about 10 s, sympy and a Monte Carlo) do not run on import; `Listas/lista-aulas-01-02-check.py` has the symbolic twin `finite_share`. The companions' independent check is `companions/check_companions.py`.

Map: L04-00-map · Formulas: L04-10-formulas

---

0 · Setup and the reused solver

```
import ast
import itertools
import math
from pathlib import Path

import numpy as np

ROOT = Path.cwd()           # run from the repository root
while not (ROOT / "Aula").exists() and ROOT != ROOT.parent:
    ROOT = ROOT.parent


def load_function(path, name):
    """Load top-level function `name` from `path` without executing the rest of the
     ↪  module."""
    tree = ast.parse(Path(path).read_text(encoding="utf-8"))
    node = next(n for n in tree.body if isinstance(n, ast.FunctionDef) and n.name ==
 ↪  name)
    ns = {}
    exec(compile(ast.Module([node], []), str(path), "exec"), ns)
    return ns[name]


# proposer_share(T, d_first, d_second): share of the round-1 proposer, T rounds, pie
 ↪  1.
proposer_share = load_function(ROOT / "Aula/notes-aulas-01-02/check_part2.py",
 ↪  "proposer_share")
```

---

1 · Multistage games: the Prisoner-Revenge Game → L04-02-carrot-and-stick

Brute force over every pure strategy profile, using the one-stage deviation principle (L04-03-one-stage-deviation): a profile is an SPE iff stage 2 is a Nash equilibrium after each of the four histories **and** no player gains by changing only his stage-1 action.

```
PD = {("M", "m"): (4, 4), ("M", "f"): (-1, 5), ("F", "m"): (5, -1), ("F", "f"): (1,
 ↪  1)}   # p. 176
REV = {("L", "l"): (0, 0), ("L", "g"): (-4, -1), ("G", "l"): (-1, -4), ("G", "g"):
 ↪  (-3, -3)}  # p. 177
HIST = list(PD)                                                  # Mm, Mf, Fm, Ff


def stage_nash(game):
    """Pure Nash equilibria of a 2x2 bimatrix given as {(a1, a2): (u1, u2)}."""
    A1 = sorted({a for a, _ in game}); A2 = sorted({b for _, b in game})
    return [(a, b) for a in A1 for b in A2
            if game[(a, b)][0] >= max(game[(x, b)][0] for x in A1)
            and game[(a, b)][1] >= max(game[(a, y)][1] for y in A2)]


def pure_spe_paths(delta):
    """Stage-1 outcomes reachable in some pure SPE, with stage 2 a pure Revenge NE
     ↪  per history."""
    ne2 = stage_nash(REV)
    paths = set()
    for plan in itertools.product(ne2, repeat=4):                # continuation NE for
     ↪  each history
        cont = dict(zip(HIST, plan))
        v = {h: tuple(PD[h][i] + delta * REV[cont[h]][i] for i in (0, 1)) for h in
 ↪  HIST}
        paths.update(stage_nash(v))                               # stage-1 one-stage
 ↪  deviations
    return paths


def two_stage_threshold(gain, carrot, stick):
    """delta* = gain / (carrot - stick); None when the stick is not a punishment."""
    return gain / (carrot - stick) if carrot > stick else None


assert stage_nash(PD) == [("F", "f")]
assert stage_nash(REV) == [("G", "g"), ("L", "l")]
assert 2 ** 5 == 32 and 32 * 32 == 1024                           # strategies, p.
 ↪  179
assert two_stage_threshold(5 - 4, 0, -3) == 1 / 3                  # (M,m), p. 183
assert two_stage_threshold(1 - (-1), 0, -3) == 2 / 3               # (F,m), p. 184
assert pure_spe_paths(0.2) == {("F", "f")}
assert pure_spe_paths(0.5) == {("M", "m"), ("F", "f")}
assert pure_spe_paths(0.7) == set(HIST)                           # all four once
 ↪  delta >= 2/3
assert ("M", "m") in pure_spe_paths(1 / 3) and ("F", "m") in pure_spe_paths(2 / 3)
# the mixed Revenge equilibrium: q = 1/2 makes L and G indifferent, value -2 (p. 177)
q = 0.5
assert 0 * q - 4 * (1 - q) == -1 * q - 3 * (1 - q) == -2
```

2 · Two-stage applications → L04-04-two-stage-applications

```
def bank_run_reduced(D, r, R):
    """Date-1 game after substituting date 2's unique NE (R, R); Gibbons Fig. 2.2.1,
     ↪  p. 75."""
    assert D > r > D / 2 and R > D
    return {("w", "w"): (r, r), ("w", "d"): (D, 2 * r - D),
            ("d", "w"): (2 * r - D, D), ("d", "d"): (R, R)}


def tariff_welfare(ti, tj, A=1.0):
    """W_i*(t_i, t_j), Gibbons p. 78, with A = a - c."""
    return (2 * A - ti) ** 2 / 18 + (A + ti) ** 2 / 9 + (A - 2 * tj) ** 2 / 9 + ti *
     ↪  (A - 2 * ti) / 3


def tournament_prize(sigma, gprime_at_opt=1.0):
    """Optimal spread w_H - w_L solving (w_H - w_L) * int f^2 = g'(e*) = 1, normal
     ↪  noise."""
    eps = np.linspace(-12 * sigma, 12 * sigma, 200_001)
    f = np.exp(-eps ** 2 / (2 * sigma ** 2)) / (sigma * math.sqrt(2 * math.pi))
    int_f2 = np.sum(f ** 2) * (eps[1] - eps[0])
    return gprime_at_opt / int_f2, int_f2


assert stage_nash(bank_run_reduced(100, 75, 150)) == [("d", "d"), ("w", "w")]   # run
 ↪  and no run
grid = np.linspace(0, 1, 100_001)
for tj in (0.0, 1 / 3, 0.5):                                          # dominant tariff (a
 ↪  - c)/3
    assert abs(grid[np.argmax(tariff_welfare(grid, tj))] - 1 / 3) < 1e-4
cells = [round(162 * tariff_welfare(a, b)) for a, b in [(0, 0), (1/3, 0), (0, 1/3),
 ↪  (1/3, 1/3)]]
assert cells == [72, 81, 56, 65]                                      # a Prisoners'
 ↪  Dilemma
assert abs((1 + 1/3) / 3 - 4 / 9) < 1e-12 and abs((1 - 2/3) / 3 - 1 / 9) < 1e-12   #
 ↪  h*, e*
prize, int_f2 = tournament_prize(1.0)
assert abs(int_f2 - 1 / (2 * math.sqrt(math.pi))) < 1e-9          # int f^2 = 1/(2
 ↪  sigma sqrt(pi))
assert abs(prize - 2 * math.sqrt(math.pi)) < 1e-6                 # about 3.545
```

3 · Alternating offers → L04-05-ultimatum-game, L04-06-finite-alternating-offers, L04-07-rubinstein-infinite-horizon

```
def finite_share_closed(T, d):
    """(11.1) and its even-T twin: x_1 = (1 - (-d)^T) / (1 + d)."""
```

```
    return (1 - (-d) ** T) / (1 + d)


def rubinstein(d1, d2):
    """Infinite-horizon share of the first proposer; None if delay is free for
    ↪ both."""
    return (1 - d2) / (1 - d1 * d2) if d1 * d2 < 1 else None


def ultimatum_best_offer(grid_step):
    """Proposer's best share on a grid when the responder accepts any positive share
    ↪ (MWG fn. 16)."""
    return 1 - grid_step


assert proposer_share(1, 0.6, 0.6) == 1                              # Prop. 11.2
assert ultimatum_best_offer(0.01) == 0.99                             # one penny to the
↪ responder
for d in (0.0, 0.3, 0.6, 0.9, 1.0):
    for T in range(1, 25):
        assert abs(proposer_share(T, d, d) - finite_share_closed(T, d)) < 1e-12
assert np.allclose([proposer_share(T, .6, .6) for T in range(1, 7)],
                   [1, .4, .76, .544, .6736, .59584])
assert proposer_share(2, 0.6, 0.6) == 1 - 0.6                        # two rounds, p.
↪ 225
assert proposer_share(39, 1, 1) == 1 and proposer_share(40, 1, 1) == 0   # delta = 1:
↪ last word wins
assert abs(rubinstein(.6, .6) - 0.625) < 1e-12                       # (11.4)-(11.5)
assert abs(proposer_share(401, .6, .6) - rubinstein(.6, .6)) < 1e-12   # (11.2)
# Gibbons p. 71: the fixed point of f(s) = 1 - d + d^2 s
d, s = 0.6, 0.0
for _ in range(200):
    s = 1 - d + d * d * s
assert abs(s - 1 / (1 + d)) < 1e-12
# different patience: three rounds and infinite horizon (check_part2.py's numbers)
assert abs(proposer_share(3, .9, .7) - 0.93) < 1e-12
assert abs(rubinstein(.9, .7) - 0.3 / 0.37) < 1e-12
h = 1e-6                                                              # signs of the
↪ derivatives
assert rubinstein(.9 + h, .7) > rubinstein(.9, .7) > rubinstein(.9, .7 + h)
# frequent offers: delta_i = exp(-r_i Delta) -> share r2 / (r1 + r2)
r1, r2 = 0.05, 0.10
assert abs(rubinstein(math.exp(-r1 * 1e-6), math.exp(-r2 * 1e-6)) - r2 / (r1 + r2)) <
↪ 1e-5
assert rubinstein(1, 1) is None
```

---

4 · Legislative bargaining → L04-08-legislative-bargaining

```
def closed_rule_k(N, d):
```

```
    """(11.8): proposer's share under the closed rule."""
    return 1 - d * (N - 1) / (2 * N)


def open_guaranteed(d):
    """(11.9)."""
    return 1 / (1 + 2 * d)


def open_risky(d):
    """Solve (11.10)-(11.12) as a linear system in (v_k, v_0, k); compare with
     ↪  (11.13)."""
    M = np.array([[1, -d / 2, -0.5],      # v_k - (d/2) v_0 - k/2 = 0
                  [-d / 2, 1, 0],         # v_0 - (d/2) v_k = 0
                  [d, 0, 1]])             # k + d v_k = 1
    v_k, v_0, k = np.linalg.solve(M, [0, 0, 1])
    return v_k, k


assert abs(closed_rule_k(3, 0.9) - 0.7) < 1e-12 and abs(closed_rule_k(5, 0.9) - 0.64)
 ↪  < 1e-12
N, d = 3, 0.9                                                  # v = 1/N solves
 ↪  (11.6)-(11.7)
v = 1 / N
k = 1 - (N - 1) / 2 * d * v
assert abs(v - (k / N + (N - 1) / (2 * N) * d * v)) < 1e-12
for d in np.linspace(0.01, 0.99, 99):
    v_k, k = open_risky(d)
    assert abs(v_k - 2 / (4 + 2 * d - d * d)) < 1e-12 and abs(k - (4 - d * d) / (4 +
     ↪  2 * d - d * d)) < 1e-12
    assert closed_rule_k(3, d) > max(open_guaranteed(d), v_k)    # footnotes 7-8
lo, hi = 0.0, 1.0                                               # where risky
 ↪  overtakes guaranteed
for _ in range(60):
    mid = (lo + hi) / 2
    lo, hi = (mid, hi) if open_risky(mid)[0] < open_guaranteed(mid) else (lo, mid)
assert abs(lo - (math.sqrt(3) - 1)) < 1e-12
```

---

5 · Bargaining over a sale → L04-09-bargaining-private-information

```
def ic_price_bounds(pS0=0.8, pB3=0.8):
    """M&R pp. 141-142: price p (seller 0, buyer 3) needed for truthful reports."""
    seller_min = (pB3 * 2 - (1 - pB3) * 1) / pB3            # (1-pB3)*1 + pB3*p >=
 ↪  pB3*2
    buyer_max = 3 - ((pS0 * (3 - 1)) - (1 - pS0) * (3 - 2)) / pS0   # (1-pS0)(1) +
 ↪  pS0(3-p) >= pS0*2
    return seller_min, buyer_max
```

```
lo_p, hi_p = ic_price_bounds()
assert abs(lo_p - 1.75) < 1e-12 and abs(hi_p - 1.25) < 1e-12 and lo_p > hi_p   # no
 ↪  price works
assert abs(0.2 * 0.2 - 0.04) < 1e-15                                  # no-trade
 ↪  efficient 4% of the time
print("Lecture 4 reference implementation: all checks pass")
```

Links

- Figures use the same expressions: `figures/make_figures.py` (run `python Notas/L04/figures/make_figures`
- Companion checks: `python Notas/L04/companions/check_companions.py` (node required).
- Earlier code: `Aula/notes-aulas-01-02/check_part2.py`, `Listas/lista-aulas-01-02-check.py`.